

Treatment of sports injuries

with BEN CLARSEN

Healing is a matter of time, but is sometimes also a matter of opportunity.

Hippocrates (c.460–370 BCE)

This chapter is integrally linked to the following chapter on the principles of injury rehabilitation. While the terms *treatment* and *rehabilitation* are often used synonymously, we define *treatment* as specific interventions aimed at tissue pathology, impairments or symptoms. Examples of treatments include targeted tendon loading (therapeutic exercise), mobilisation of a cervical spine facet joint (manual therapy), extra-corporal shockwave therapy (electrotherapy) and non-steroidal anti-inflammatory drugs (therapeutic medications). In contrast, we define *rehabilitation* as a holistic process aimed at restoring the athlete to his or her pre-injury function—a part of which involves the application of the right treatments at the right time.

This chapter provides essential background for the treatments detailed in Part B: Regional Problems. Evidence for treatment effectiveness is continually changing. However, remember that our craft remains as much art as science. Level 1 evidence is not always available (see box) and the decision to use (or not to use) certain treatments is also influenced by our experience, professional training and patient's expectations (see 'A cautionary tale' below and Chapters 1 and 2). Nevertheless, clinicians should have a solid understanding of the theoretic rationale for the treatments they provide, as well as up-to-date knowledge of the evidence of their effectiveness.

The effect of each treatment should be evaluated by objectively comparing symptoms and signs before and after the treatment, and at the next visit. Functional testing and patient-reported outcome measures (PROMs—Chapter 16) should also be used regularly throughout the course of treatment. This enables the clinician to choose

the most appropriate mode of treatment for the specific injury and individual. In presentations that fail to improve, this also allows the clinician to change modalities or pursue further investigations.

High-level evidence is not always available

There are no randomised controlled studies to demonstrate that jumping out of a plane with a parachute leads to superior outcomes than jumping without one.¹ The best evidence for the effectiveness of parachutes comes from retrospective case series (level 4 evidence; Fig. 17.1). Would this stop you from recommending their use?



Figure 17.1 The level of research evidence supporting the use of parachutes is low CORBIS

A cautionary tale

with KIERAN O'SULLIVAN

Before we launch into specific treatments and the evidence that underpins them, we need to share a cautionary tale. Just as Chad Cook (Chapter 14) alerted us that clinical assessment is not as accurate as the 1990s textbooks made out, treatment is not what a glitzy pharmaceutical video might suggest it is.

Unfortunately, several very important factors—that are unrelated to the pathology of a condition itself—can influence treatment outcomes. You have seen patients who are positive in their outlook and others who seem to live in a world of gloom and doom. You probably suspect that this might affect the outcome of treatment, and evidence indicates it most certainly does.

Patients' response to treatment is greatly influenced by *cognitive* (e.g. thoughts, self-efficacy, hypervigilance), *psychological* (e.g. depression, stress, anxiety, perceived injustice), *social* (e.g. work, family, financial, cultural, social support, life events) and *lifestyle* (e.g. poor sleep, sedentary behaviours, physical activity levels, diet, smoking) factors (Fig. 17.2). These systemic factors are widely acknowledged as being key in the onset and maintenance of pain^{2–4} among non-sporting populations reporting pain or injury (e.g. chronic low back pain, hip and knee osteoarthritis). Sports medicine is just beginning to embrace this undeniably important concept.

PRACTICE PEARL

Psychosocial, cognitive and lifestyle factors are related to (i) the onset of symptoms,^{5,6} (ii) treatment response^{7,8} and (iii) return to sport.^{9,10}

While these systemic factors are often perceived as difficult to modify, they are likely to be at least as modifiable as many physical factors (e.g. posture, muscle thickness), which are often the main targets for treatment, but which often do not objectively change as the athlete recovers.^{11, 12} Successful clinical outcomes after treatments typically considered to be 'physical', such as exercise, are often mediated by changes in cognitive and psychological factors.¹³ Critically, there is evidence that these systemic factors are related to more traditional physical factors (e.g. fear affects movement patterns and, thus, can slow the athlete's return to sport).^{14, 15}

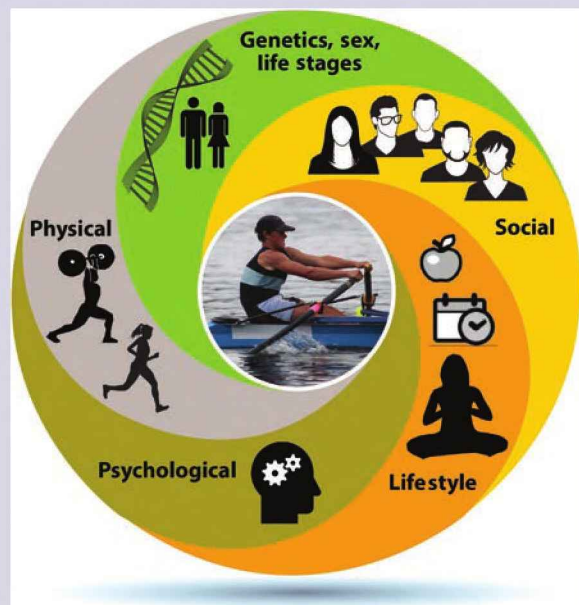


Figure 17.2 A wide range of factors influence patients' response to treatment

PRACTICE PEARL

Rather than considering rehabilitation from injury as physical or psychological, rehabilitation might often need to combine these aspects in a manner that best matches the individual needs of each athlete.

For example, as discussed below, exercise is a core component of injury rehabilitation. Since exercise can enhance a range of systemic factors (e.g. sleep, mood, catastrophising, fear, self-efficacy), there may be occasions when athletes benefit more from the health-enhancing systemic effects of exercise (e.g. it aids relaxation and sleep) than from local loading tissue effects (e.g. an athlete whose pain or injury is not explained by the typical tissue injury model). In situations such as this, clear communication with athletes regarding the mechanism of effect of exercise, and the context in which that exercise is completed, is critical. Furthermore, traditional rehabilitation can then evolve to incorporate additional treatment components (e.g. relaxation, cognitive behavioural therapy, sleep hygiene, dietary advice) to influence several of these wellness factors as required with each athlete.

THERAPEUTIC EXERCISE

The health benefits of regular physical activity have been recognised since before the time of Hippocrates and proven more and more convincingly since Jeremy Morris' London bus study in the 1950s. The slogan 'exercise is medicine' has been adopted by many public health groups aiming to prevent non-communicable diseases such as cardiac disease, type 2 diabetes and cancer.

'Exercise is medicine' also applies to the treatment of musculoskeletal injury. There is abundant evidence for exercised-based therapy as treatment of a wide range of conditions including tendinopathy,¹⁶ osteoarthritis^{17, 18} and lower back pain.¹⁹

The following section will review a number of mechanisms that may explain the benefits of therapeutic exercise, such as stimulation of tissue repair and remodelling, altering biomechanics and exercise-induced hypoalgesia.

PRACTICE PEARL

Abundant high-level evidence supports therapeutic exercise as a first-line treatment across a wide range of conditions. It should be considered as a first-line treatment for almost any musculoskeletal condition.

Stimulation of repair and remodelling: mechanotherapy

The prescription of therapeutic exercise to promote beneficial tissue adaptations has been called 'mechanotherapy'.²⁰ This refers to the employment of mechanotransduction—the process by which cells convert physiological mechanical stimuli into biochemical responses—to promote repair and remodelling of injured tissue (see boxed text).

Mechanotransduction: how loading tissue can stimulate repair and remodelling

The process of mechanotransduction is integral to the homeostatic maintenance of the musculoskeletal system in the absence of injury. It also explains how therapeutic exercise can strongly influence injured tissue.

Mechanotransduction consists of three steps:

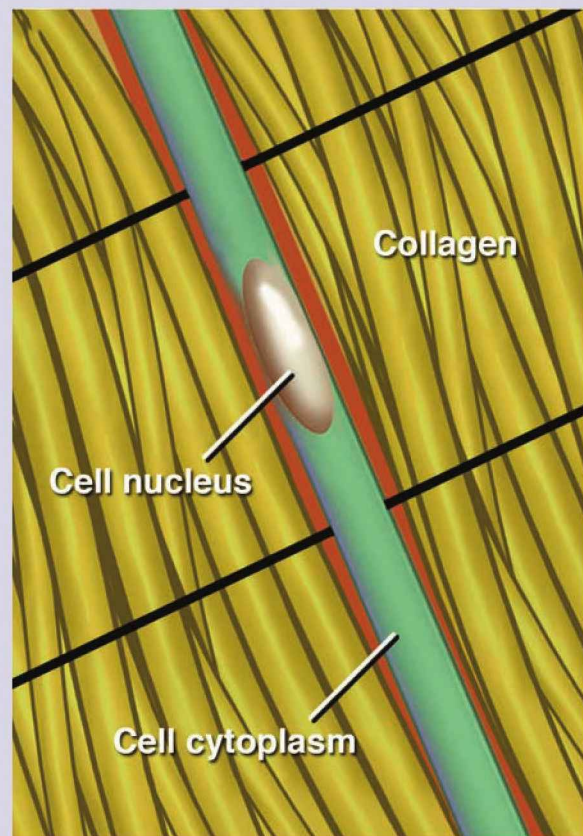
(i) mechanocoupling, (ii) cell–cell communication and (iii) the effector response. Here we use tendon tissue to illustrate this process, but the fundamental principles also apply to other musculoskeletal tissues such as bone, muscle, ligament and articular cartilage.

Mechanocoupling—the loading trigger

Mechanocoupling refers to physical load (often shear or compression) causing a physical perturbation to cells that make up a tissue. For example, with every step, the Achilles tendon receives tensile loads generated by the three elements of the gastrocnemius–soleus complex and, thus, the cells that make up the tendon (Fig 17.3a) experience tensile and shearing forces (Fig. 17.3b). Tendon cells can also experience compression forces (Fig. 17.3c). These forces elicit a deformation of the cell that can trigger a wide array of responses depending on the type, magnitude and duration of loading.²¹ The key to mechanocoupling, as the name suggests, is the direct or indirect physical perturbation of the cell, which is transformed into a variety of chemical signals both within and among cells.

Cell–cell communication—distributing the loading message

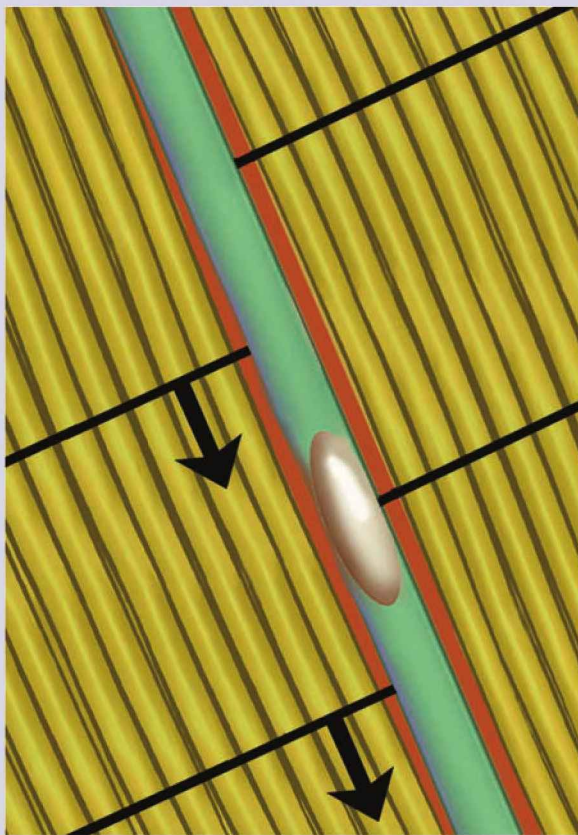
The previous paragraph illustrated mechanocoupling by focusing on a single cell, but let us draw back to examine a larger tissue area that contains thousands of cells embedded



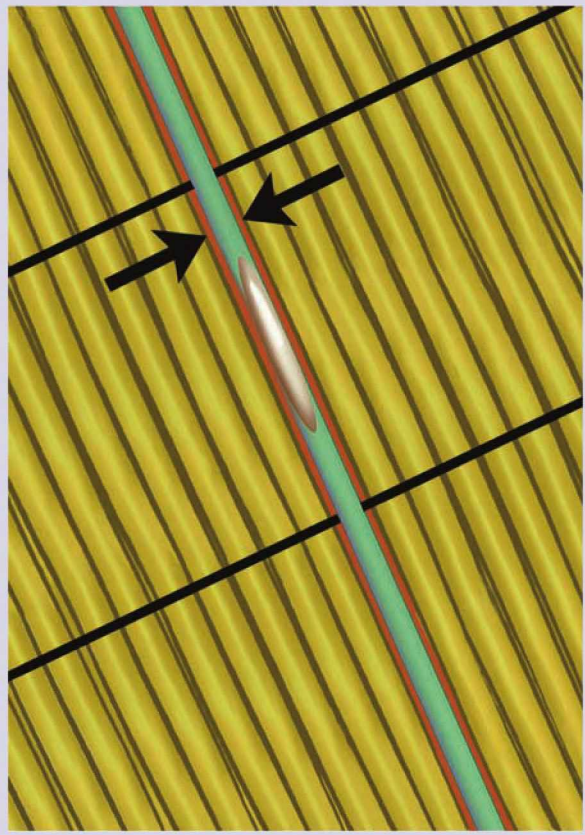
(a)

Figure 17.3 The various elements of the tendon cell (a)

continued



(b)



(c)

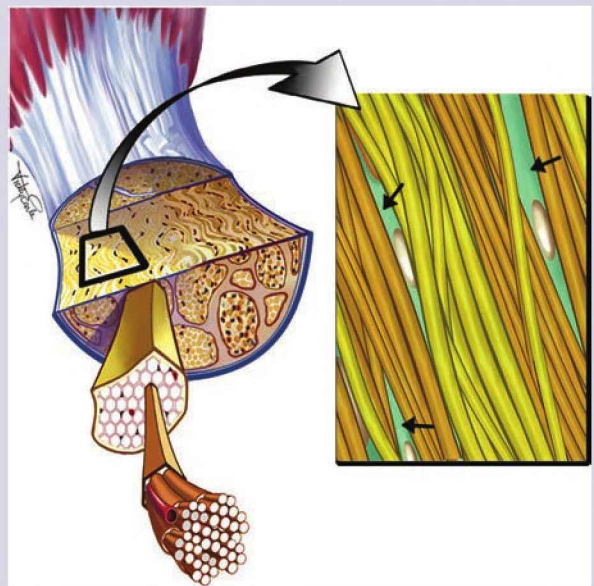
Figure 17.3 (cont.) The various elements of the tendon cell undergo shear (b) and compression (c) during a loading cycle

within an extracellular matrix (Fig. 17.4). The signalling proteins for this step include calcium and inositol triphosphate (IP_3). The process of cell–cell communication is best understood by illustration (Fig. 17.4). The critical point is that stimulus in one location (location 1 in Fig. 17.4c) leads to a distant cell registering a new signal (location 2 in Fig. 17.4e) even though the distant cell does not receive a mechanical stimulus.²¹

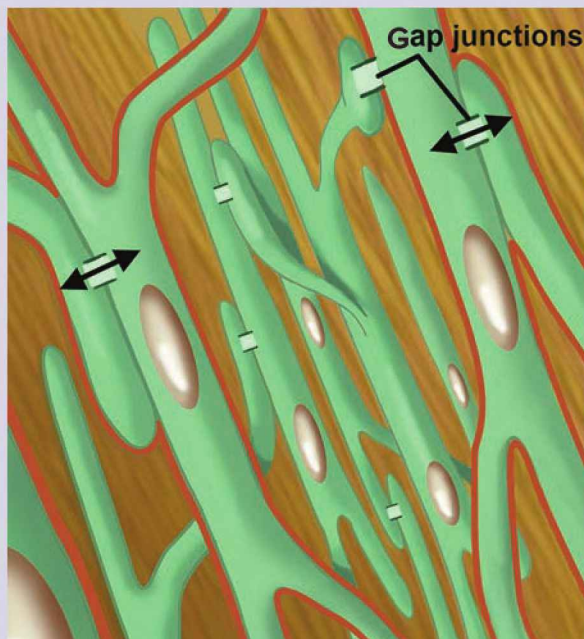
Effector cell response

To illustrate the third part of mechanotransduction (effector cell response), we focus on a single cell at the boundary of the extracellular matrix (Fig. 17.5). With movement, integrin proteins located in the cell membrane activate at least two distinct pathways that influence the cell nucleus to initiate the production of new proteins. These pathways include physical and biochemical signals (Fig. 17.5d).

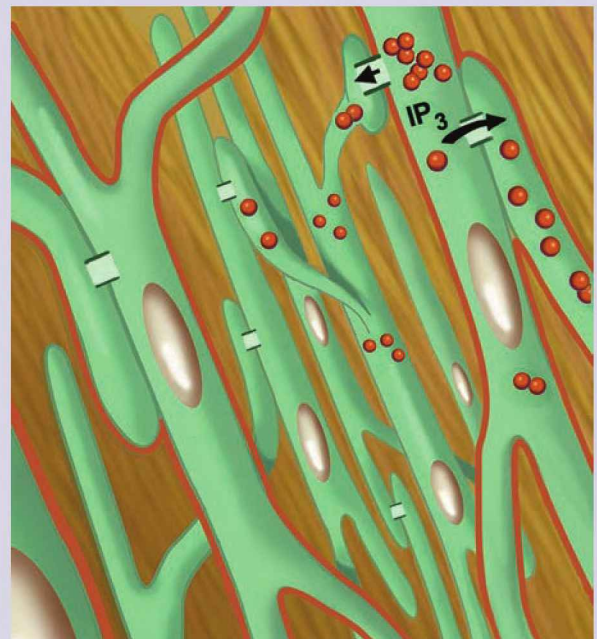
Figure 17.4 Tendon tissue provides an example of cell–cell communication (a) The intact tendon consists of extracellular matrix (including collagen) and specialised tendon cells (arrowheads)



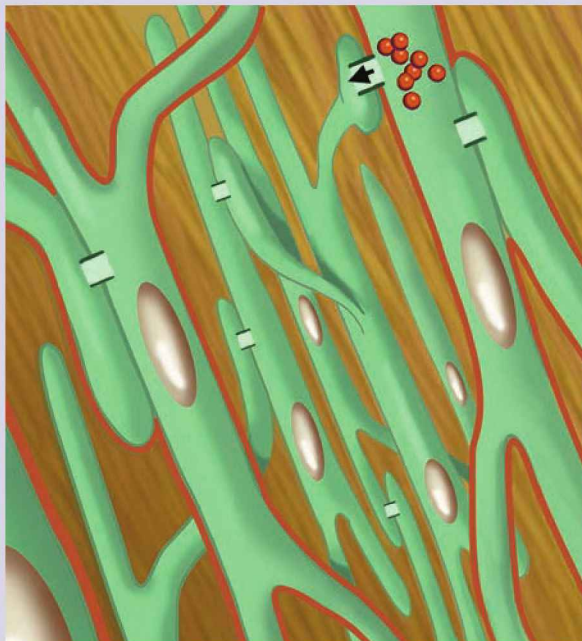
(a)



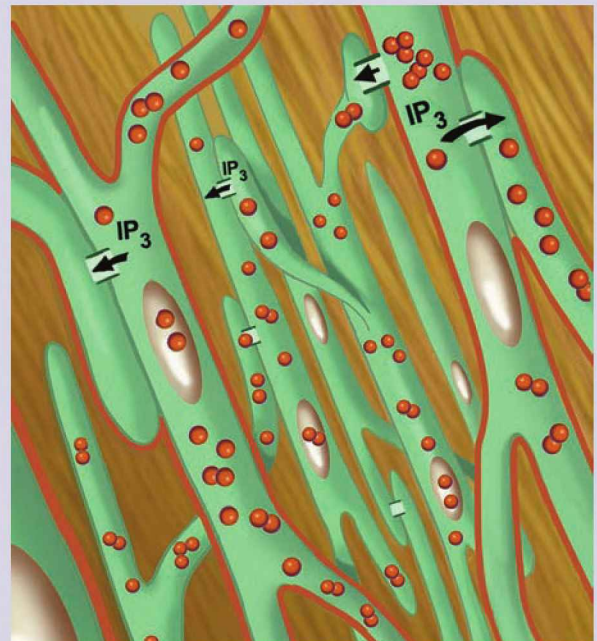
(b)



(d)



(c)

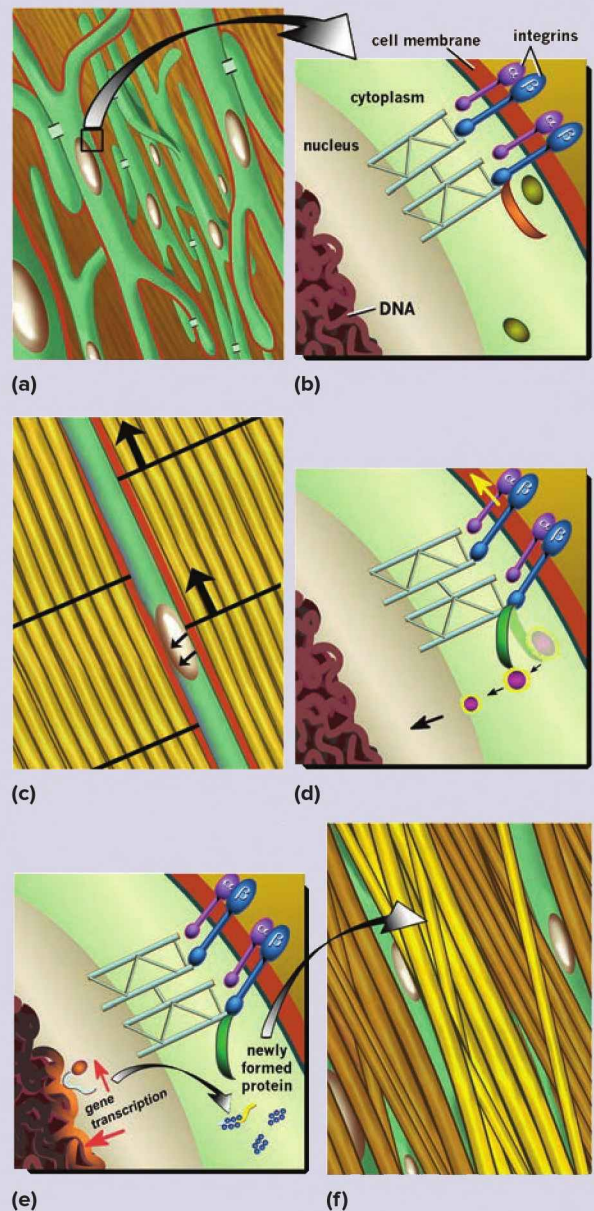


(e)

Figure 17.4 (cont.) (b) Tendon with collagen removed to reveal the interconnecting cell network. Cells are physically in contact throughout the tendon, facilitating cell-cell communication. Gap junctions are the specialised regions where cells connect and communicate small charged particles. They can be identified by their specific protein, connexin 43. Time course of cell-cell communication from (c) the beginning, through (d) the midpoint to (e) the end. The signalling proteins for this step include calcium (red spheres) and inositol triphosphate (IP₃)

continued

Figure 17.5 Mechanical loading stimulates cells to synthesise protein (a) A larger-scale image of the tendon cell network for orientation. We focus on one very small region (b) Zooming in on this region reveals the cell membrane, the integrin proteins that bridge the intracellular and extracellular regions, and the cytoskeleton, which functions to maintain cell integrity and distribute mechanical load. The cell nucleus and the DNA are also illustrated (c) With movement (shearing is illustrated), the integrin proteins activate at least two distinct pathways (d) One pathway involves the cytoskeleton that is in direct physical communication with the nucleus (yellow arrow) (i.e. tugging the cytoskeleton sends a physical signal to the cell nucleus). Another pathway is triggered by integrins activating a series of biochemical signalling agents, which are illustrated schematically (green lever). After a series of intermediate steps, those biochemical signals also influence gene expression in the nucleus (e) Once the cell nucleus receives the appropriate signals, normal cellular processes are engaged. mRNA is transcribed and shuttled to the endoplasmic reticulum in the cell cytoplasm, where it is translated into protein. The protein is secreted and incorporated into the extracellular matrix (f) In sum, the mechanical stimulus on the outside of the cell promotes intracellular processes leading to matrix remodelling



A range of studies have shown or implied a potential for mechanotherapy to promote healing of muscle, tendon, ligament, cartilage and bone. The key to successful clinical application of mechanotherapy is most often finding the correct dosage (and thus progression of exercises).

In some cases, such as patellar and Achilles tendinopathy, literature exists to guide the clinician's exercise prescription. 'Research-proven' treatment regimens, such as tendon training protocols (Chapter 40) or 'heavy slow resistance training'²² have considerable clinical popularity and are

often applied to a variety of other injury types. However, we warn against taking a 'one size fits all' approach. General training principles of progressive overload, periodisation and specificity should be followed (Chapter 10), and exercise prescription should be individualised according to each patient's functional capacity, sport-specific demands and treatment response.

The following section reviews the specific mechanisms and clinical application of mechanotherapy for muscle, tendon, cartilage and bone.

Muscle

Muscle offers one of the best opportunities to exploit and study the effects of mechanotherapy as muscle is highly responsive to changes in loading. Increasing load on muscle leads to the immediate, local upregulation of mechanogrowth factor (MGF), which in turn stimulates muscle hypertrophy via activation of satellite cells.²³ The opposite mechanism is at play in cases where athletes are forced to rest a body part (immobilise or greatly reduce muscle load), such as after a fracture or surgery. In such cases, mechanotransduction causes catabolism of tissue—the tissue adapts to the load it senses.

The benefits of loading include improved alignment of regenerating myotubes, faster and more complete regeneration, and minimisation of atrophy of surrounding myotubes.²⁴ We revisit this topic later in the section on acute injury management.

Tendon

Tendon is a dynamic, mechanoresponsive tissue. One of the major load-induced responses in tendon, which has been shown both *in vitro*²⁵ and *in vivo*,^{26–28} is an upregulation of insulin-like growth factor (IGF-I). This upregulation of IGF-I is associated with cellular proliferation and matrix remodelling within the tendon. However, other growth factors and cytokines in addition to IGF-I are also likely to play a role.²⁹

Recent studies show a significant increase in stiffness, Young's modulus and cross-sectional area in human tendons subjected to mechanical loading programmes.³⁰ However, the extent of tissue adaptation appears to be highly dependent on the applied loading parameters. In particular, several studies indicate that high-intensity muscle contractions (70–90% of maximum) are necessary to increase tendon stiffness,^{31–34} and that interventions longer than 12 weeks are necessary to achieve optimal adaptations.^{35, 36} The mechanical, material and morphological properties of tendons respond similarly to high-intensity loading programmes involving eccentric, concentric-eccentric and isometric muscle contractions.^{16, 30, 37} However, plyometric training may not lead to optimal adaptations.³⁰

Articular cartilage

Like other musculoskeletal tissues, articular cartilage is populated by mechanosensitive cells (chondrocytes), which signal via pathways highly comparable to those described for tendon. Animal studies have shown that immobilisation leads to cartilage atrophy and loss of stiffness, whereas cartilage subjected to high loads has a higher proteoglycan content and higher cell volume, and is stiffer.³⁸

Recent advances in magnetic resonance imaging (MRI) technology, in particular the ability to estimate cartilage glycosaminoglycan (GAG) content,³⁹ have improved our understanding of loading on human articular cartilage. A 16-week moderate loading program increased the GAG content of articular cartilage among patients at high risk of developing tibiofemoral osteoarthritis.⁴⁰ Vigorous loading programs have provided mixed results. Some led to improved cartilage quality among people with⁴¹ and without⁴² osteoarthritic changes, whereas others were detrimental in people with pre-existing bone marrow lesions.⁴³

There is little high-level evidence to guide loading dosage for articular cartilage injuries—the dose-response curves are still being determined. However, there was clinical improvement among 48 patients with full-thickness femoral condyle lesions who utilised a 3-month progressive exercise program consisting of moderate resistance training (3–4 sets of 8–10 repetitions at 40–60% of maximum), as well as neuromuscular and cardiovascular training.⁴⁴ A key clinical principle of loading articular cartilage lesions is to avoid pain and swelling during or after training.^{44, 45}

Bone

The process of mechanotransduction has been well described in human bone.⁴⁶ Appropriate loading of osteocytes, which are believed to be the primary mechanosensors, activates a number of extracellular and intracellular signalling pathways which stimulate osteoblast proliferation and new bone formation. However, both insufficient and excessive loading of bone stimulates osteoclast activity, which leads to a loss of bone tissue.⁴⁶

There is growing evidence that mechanotransduction can be exploited to improve fracture healing, through either active load bearing (e.g. partial weight-bearing mobilisation), pneumatic compression or electrophysical agents.⁴⁷ There are no definitive guidelines on the amount of loading required to optimise bone formation, particularly as many factors influence each case, such as the loading type, the site and size of the fracture, and the patient's age and hormonal status. However, studies agree that dynamically applied loads are more effective than static loads in stimulating bone remodelling.⁴⁷

Altering biomechanics: motor-control training

In addition to stimulating repair and remodelling of injured tissues, clinicians also prescribe therapeutic exercise to influence the way a patient moves his or her body. We often refer to this as 'neuromuscular' or 'motor-control' training.

Clinicians usually prescribe motor-control training in an attempt to:

1. Directly or indirectly unload injured or sensitised tissue. For example, overhead athletes with rotator cuff tendinopathy are often prescribed scapular retraction and upward rotation exercises to reduce posterior impingement of their cuff tendons during throwing (Chapter 24).
2. Reduce the risk of injury or re-injury by improving strength, balance, coordination and/or sensorimotor control. For example, patients recovering from anterior cruciate ligament injury are prescribed a wide range of neuromuscular exercises throughout the entire rehabilitation process to optimise their functional outcome and reduce the risk of re-injury (Chapter 35).

There are no definitive guidelines for the dosage of motor control training. However, neuromuscular 'perturbation' programs performed three times per week over a period of 3–4 weeks have been shown to alter motor control patterns in uninjured⁴⁸ and ACL-injured athletes,⁴⁹ and to improve ACL rehabilitation outcomes.⁵⁰ Similarly, training programmes aiming to improve neuromuscular control during standing, cutting, jumping and landing have been shown to prevent injuries in female handball and football players when performed for 15 minutes, three times per week (Chapter 12).^{51, 52} As for all types of skill acquisition, regular high-quality practice is necessary to learn and engrain new motor patterns.

PRACTICE PEARL

Effective motor control training programmes focus on movement quality and are sport-specific. Athletes need to concentrate hard throughout the training session and should understand what they are trying to achieve and why it is important. When introducing a new exercise, the clinician should provide a lot of feedback; however, athletes should always be encouraged to assess their own performance.

Motor control training programmes should target specific deficits identified during the clinical assessment. Complex movements should first be broken down into separate components, before the whole task is trained. As pain (or the distraction caused by pain) may interfere with the normal neuroplastic changes that occur with skill training,⁵³ motor-control training should be pain-free.

Exercise-induced hypoalgesia

In addition to stimulating local repair and remodelling and altering biomechanics, clinicians may prescribe therapeutic exercise due to its hypoalgesic effects. There is a large body of evidence demonstrating that in healthy adults and a range of patient populations, exercise attenuates experimentally-induced pain.⁵⁴ This is achieved by elevating pain thresholds and pain tolerance, as well as by reducing pain intensities during and after exercise.^{55, 56}

Exercise-induced hypoalgesia is likely to be caused by a combination of factors, including the release of endogenous opioids and other non-opioid substances affecting pain perception such as endocannabinoids, serotonin and norepinephrine (noradrenaline),^{56, 57} as well as activation of spinal or supraspinal mechanisms of pain inhibition.^{58, 59} Different exercise parameters may alter the predominant mechanism.^{56, 60}

Exercise type and dose to maximise hypoalgesia

Aerobic exercise, isometric muscle contractions and traditional resistance training have all been shown to reduce pain, with isometric exercise having the largest effect.⁵⁴ Although a recent meta-analysis was unable to calculate the exact dosage required to maximise exercise-induced analgesia, the dose-response relationship favours moderate to high intensities for both aerobic and isometric exercise.⁵⁴ Current recommendations, therefore, are that aerobic exercise be performed at intensities above 75% of maximal aerobic capacity (VO₂max) for at least 10 minutes,⁵⁴ and isometric contractions should be above 50% of maximum voluntary contraction (MVC) held for long durations or until task failure.^{61, 62} Recent studies of athletes with patellar tendinopathy have found that five 45-second isometric contractions at 70% of MVC with two minutes' rest between contractions significantly reduced pain for at least 45 minutes after training.⁶³ Based on these findings, isometric training is currently recommended in the early management of tendinopathy (Chapter 4).

In certain groups of patients with generalised chronic pain such as fibromyalgia syndrome, vigorous-intensity exercise has been shown to increase pain. However, low- to moderate-intensity aerobic and isometric exercise has an effective hypoalgesic effect in this group.⁵⁴

PRACTICE PEARL

As the hypoalgesic effects of exercise are transient, regular exercise is necessary to reduce pain effectively. For example, isometric contractions for the treatment of painful tendinopathy should be performed several times every day.

ACUTE INJURY MANAGEMENT

Acute soft tissue injury causes an immediate inflammatory response that manifests clinically as pain, redness, swelling and loss of function. Inflammation is an essential process that initiates tissue regeneration and repair. However, excessive bleeding and oedema can delay recovery and may cause secondary ischaemic damage to nearby tissues.⁶⁴ The goal of acute injury management, therefore, is to prevent excessive bleeding and oedema, reduce pain and promote faster return to function.

The principles of acute injury management are based largely on expert opinion and laboratory studies (level 4–5 evidence), and there are few randomised controlled trials upon which to base recommendations.^{65, 66} Traditionally, treatment is guided by one of the most commonly known acronyms in sports medicine: PRICE (**P**rotection, **R**est, **I**ce, **C**ompression and **E**levation). Recently, however, a call has been made to change the acronym to POLICE:⁶⁷

- P** Protection
- OL** Optimal loading
- I** Ice
- C** Compression
- E** Elevation

Protection

A brief period of complete immobilisation is necessary to prevent excessive distention at the injury site, reduce the size of the haematoma and thus minimise the size of the connective tissue scar.⁶⁸ During this period, the clinician may consider the use of casts, splints or rigid braces to protect the injured body part, as well as crutches for severe lower-limb injuries.

Although protection is important in the period immediately following trauma, prolonged immobilisation has a detrimental effect on healing in a range of soft tissues, including ligament, tendon and muscle.^{68–71} Immobilisation also affects surrounding structures, leading to joint stiffness, degenerative changes in articular cartilage, osteopenia, muscle atrophy and loss of strength. It is therefore important that tissues remain completely unloaded for as short a period as possible. This will vary depending on the extent and nature of the injury and the tissue involved. For most muscle injuries, two days is sufficient before the forming scar has adequate tensile strength to withstand gentle stress.²⁴ Severe ligament sprains (grades II and III) may require up to ten days of immobilisation before the patient can begin controlled loading.^{72, 73}

Optimal loading

Optimal loading replaces *rest* in the acute management acronym due to concerns that the word *rest* may encourage an overly conservative approach that fails

to harness the benefits of early tissue loading through exercise.^{67, 74}

In comparison to immobilisation, early loading has been shown to have a range of beneficial strength, morphological and functional effects on regenerating tissue, and on the wider neuromusculoskeletal system.^{67, 75} For example, early mobilisation of lateral ankle sprains has been shown to improve subjective function, patient satisfaction, swelling, and return to activity, work and sports participation (level 1 evidence).^{76, 77} Basic science studies also support early loading compared to immobilisation for a range of soft tissue injuries.^{68–71} For example, early mobilisation of muscle injuries improves capillarisation and muscle fibre regeneration, and leads to a more parallel orientation of the regenerating myofibres compared to immobilisation (Fig. 17.6).⁶⁸ Muscle injuries demonstrate persisting weakness when treated with prolonged immobilisation (Fig. 17.7).⁶⁸

A major challenge facing the clinician is to determine the 'optimal' loading for each individual at each phase of healing. This should be driven by the tissue type and pathological presentation, as well as the tissue adaptation required; for example, increased tensile strength, collagen reorganisation, increased muscle-tendon unit stiffness or neural reorganisation.⁷⁴

The clinician needs to consider a range of variables when attempting to achieve optimal loading, such as the magnitude, duration, frequency, direction, intensity and direction of load. Manipulation of these variables will not only affect local tissue healing processes, but also stimulate central nervous system adaptations leading to improved motor learning and movement efficiency.

Variable loading offers a range of potential benefits. First, small variations in the magnitude, direction and rate of loading may provide injured tissues with a degree of stress shielding from overload. Second, load variability

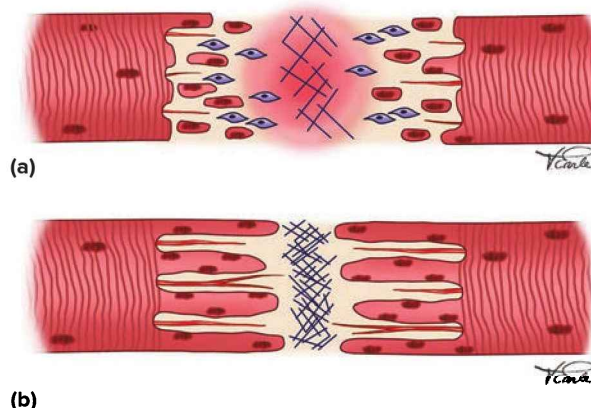


Figure 17.6 Muscle injury treated with (a) immobilisation and (b) early mobilisation. ADAPTED FROM JÄRVINEN ET AL.⁶⁸

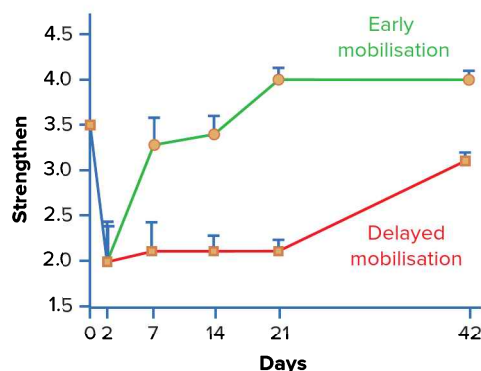


Figure 17.7 Strength improvements in rat skeletal muscle injuries treated with a progressive exercise protocol after two days of immobilisation (shown in green) and in injuries treated with the same protocol after 21 days of immobilisation (shown in red). Note that strength did not naturally return during the immobilisation period. ADAPTED FROM JÄRVINEN ET AL.⁶⁸

may provide a better mechanotransductive effect through broader stimulation of mechanoreceptors and prevention of accommodation. Finally, variable tensile, compressive and torsional forces may promote the creation of a stronger biological scaffold that is better able to withstand a range of loading types.⁷⁴

The characteristics of optimal and suboptimal loading are outlined in Table 17.1.

Ice

Ice (or other forms of cryotherapy) is one of the most common treatment modalities used in the initial management of acute musculoskeletal injuries.⁶⁵ Cryotherapy aims to decrease oedema through vasoconstriction and reduce secondary hypoxic injury by lowering the metabolic demand of injured tissues.^{65, 68} Local analgesia is thought to occur when the skin temperature drops below 15°C because of decreasing nerve conduction velocity.^{78, 79}

Laboratory studies show that cooling injured tissue to between 5°C and 15°C reduces cellular metabolism, white blood cell activity, necrosis and apoptosis.^{64, 80} However, this amount of cooling is difficult to achieve in practice, with human studies showing that intensive cooling leads to tissue temperatures of between 21°C and 25°C. The actual metabolic effects of cryotherapy may therefore be questionable, particularly for deep injuries or in patients with higher levels of adipose tissue.⁸⁰

Nevertheless, the analgesic effects of cryotherapy are well established and icing remains a part of current acute injury management guidelines.⁸¹ Ice is typically applied for 20 minutes every two hours for the first 48–72 hours following injury. Intermittent application—for example, 10 minutes on, 10 minutes off—may enhance analgesia and reduce the risk of adverse reactions such as skin burns and nerve damage.^{78, 81} Ice should not be applied where local tissue circulation is impaired (e.g. in Raynaud's phenomenon or peripheral vascular disease), or to patients who suffer from a cold allergy.

Although there are many commercial cryotherapy products available, in most situations a plastic bag containing crushed ice and water remains the modality of choice.⁸² A wet towel should be placed between the ice pack and the skin. Cryotherapy modalities are summarised in Table 17.2.

Compression

Compression is often used following acute soft tissue injury in order to increase pressure gradients in the venous and lymphatic systems, thereby counteracting oedema. Combining compression with ice may also help reduce tissue temperatures.⁸³ Although there are a number of specific products commercially available, such as cold compression wraps and devices, a standard elastic bandage is a cheap, versatile and effective solution. Compression can be focussed on certain tissues by applying ice packs underneath the elastic bandage. Similarly, pressure peaks

Table 17.1 Characteristics of optimal and suboptimal loading⁷⁴

Optimal loading	Suboptimal loading
Directed to appropriate tissues	Non-specific generalised loading
Loading through functional ranges	Loading through limited ranges of movement
Appropriate blend of compressive, tensile and shear loading	Loading exclusively in a single manner
Variability in magnitude, direction, duration and intensity	Constant, unidirectional load
Include neural overload	Minimal neural stimulus
Tailored to individual characteristics	Generic, non-individualised
Functional	Non-functional, isolated segmental loading

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Table 17.2 Cryotherapy modalities used to treat sports injuries

Modality	Description	Special concerns	Surface temperature	Duration	Exercise during application	Expense
Reusable cold packs	Durable plastic packs containing silica gel that are available in many sizes and shapes	Apply a towel between the bag and skin to avoid nerve damage or frostbite	≤15°C (59°F)	20–30 min	No	Inexpensive
Endothermal cold packs	Packets are squeezed or crushed to activate: convenient for emergency use	Single use only	20°C (68°F)	15–20 min	No	Expensive
Crushed ice bags	Crushed ice moulds easily to body parts	Apply a towel between the bag and skin to avoid nerve damage or frostbite	0°C (32°F)	5–15 min	No	Inexpensive
Vapocoolant sprays	Easily portable therapy for regional myofascial pain syndrome, acute injuries, pain relief, and in rehabilitation with spray and stretch techniques	Intermittently spray the area for <6 sec to avoid frostbite	Varies depending on duration of treatment	Multiple brief sprays	Spray <6 sec and stretch to increase range of motion	Expensive
Ice water immersion	An athlete's body part is submerged in cold water	Carries the most risk of hypersensitivity reactions; restrict amount of extremity immersion	0°C (32°F)	5–10 min	Allows motion of the extremity during treatment	Inexpensive
Ice massage	Used to produce analgesia: freeze water in a foam cup, then peel back cup to expose the ice; massage area as often as needed	Apply for short intervals to avoid frostbite; avoid excess pressure	0°C (32°F)	5–10 min	Can allow supervised, gentle, stretching during analgesia	Inexpensive
Refrigerant inflatable bladders	When cold and compression are needed	Avoid excess compression	10–25°C (50–77°F)	Depends on temperature	No	Expensive
Thermal cooling blankets	To provide constant temperature, such as after surgery	Scrutinise temperature settings	10–25°C (50–77°F)	Depends on temperature	No	Expensive
Contrast baths	Transition treatment between cold and heat for a subacute injury, sympathetic mediated pain, stiff joints	Do not use in acute setting due to potential to increase blood flow	Hot bath 40.5°C (105°F) Cold bath 15.5°C (60°F)	4 min hot, 1 min cold	Allows motion of the extremity during treatment	Inexpensive

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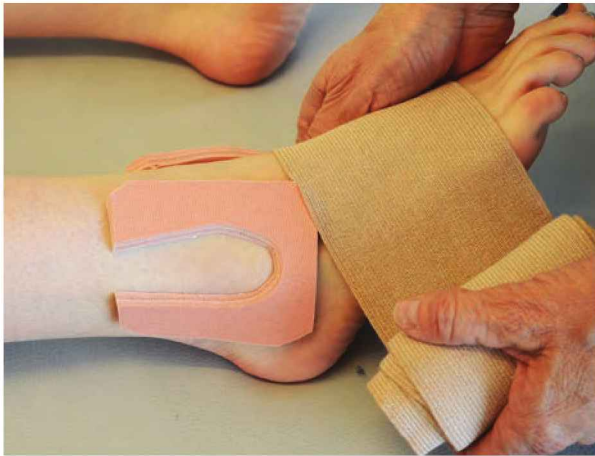


Figure 17.8 Applying compression to a lateral ankle sprain: horseshoe-shaped felt around the ankle malleolus may help distribute compression pressures more evenly

around bony prominences can be avoided by applying felt or other material to fill gaps (Fig. 17.8).

Despite a lack of consensus on the optimal compression pressures, there is some evidence that high pressures (80 mmHg) are ineffective.⁸¹ Standard protocols often recommend light to moderate pressures of 15–35 mmHg. Bandaging should start just distal to the injury site, with each layer of bandage overlapping the underlying layer by one-half. It should extend to at least a hand's breadth proximal to the injury margin.

Elevation

The rationale for elevation stems from the fundamental principles of physiology and traumatology. Elevation of an injured extremity above the level of the heart results in a decrease in hydrostatic pressure and, subsequently, reduces the accumulation of interstitial fluid.²⁴ Elevation can be achieved by using a sling for upper limb injuries and by resting lower limbs on a chair, pillows or bucket. It is important to ensure that the lower limb is above the level of the pelvis. A graduated return to standing after elevation is likely to minimise rebound swelling and discomfort.⁸¹

Do no HARM!

In the 72 hours following injury, **HARM**-ful factors should be avoided:

H Heat and heat rubs

Heat may increase the bleeding at the injured site. Avoid hot baths, showers, saunas, heat packs and heat rubs.

A Alcohol

A moderate consumption of alcohol after eccentric-based leg exercises has shown to significantly increase

the loss of dynamic and static quadriceps strength.⁸⁴ Alcohol may mask pain and severity of injury and therefore increase the risk of re-injury.⁸⁵ To minimise exercise-related losses in muscle function and to accelerate recovery, avoidance of alcohol post injury is paramount.

R Running/moderate activity

Running or any form of moderate activity can cause further damage at the injury site.

M Massage/vigorous soft tissue therapy

Vigorous massage should be avoided in the first 24–48 hours. It could cause further bleeding and swelling to the injury site.

MANUAL TREATMENTS

In this section, we refer to 'hands-on' treatments, including techniques targeting joints (mobilisation and manipulation), soft tissues (massage/myofascial release techniques) and nerves (neurodynamic techniques). There is a substantial body of evidence demonstrating immediate effects of manual treatments, such as altered pain sensation, muscle activation and joint range of motion.^{86–91} However, these effects are unlikely to be maintained over time, or to change the natural history of a disorder. Therefore, clinicians should regard manual treatments as an adjunct to addressing the underlying causes of injury.

Manual treatments are typically directed at specific tissues or structures identified as being abnormal on physical examination, such as stiff joints, tight muscles or restricted nerves (Chapter 15). Traditionally, the rationale for using manual treatments is their proposed local biomechanical and physiological effects; for example, to reduce joint stiffness, increase intramuscular temperature and circulation, or break up adhesions in connective tissue. However, the following factors suggest that local tissue changes are unlikely to explain the clinical benefits of manual treatments:

- Spinal manipulations are only associated with transient increase in joint mobility; there is no evidence of lasting changes in mobility⁹² or joint position.⁹³
- There is no evidence that massage affects passive or active muscle stiffness, and the effect on global range of motion and muscle blood flow is marginal.⁹⁴
- Clinical methods for assessing joint position, joint motion and soft tissue qualities have poor reliability.^{95–98} Different clinicians are therefore unlikely to agree upon which structures require treatment.
- The effect of manual treatments may be distant from the area treated. For example, treatment of the thoracic and cervical spine may affect lateral elbow pain (Chapter 25).^{99, 100}

The actual mechanism of manual treatment is thought to be a complex neurophysiological response from the peripheral and central nervous system, initiated by a local biomechanical stimulus.⁹⁵ Mechanical stimulation of afferent neurones located in skin, joints and muscle can activate a number of responses such as pain inhibition^{86, 101-103} and neuromuscular inhibition or facilitation.^{100, 104-106} Manual treatments also have been shown to have autonomic,¹⁰⁷⁻¹⁰⁹ endocrine,^{110, 111} immunologic¹¹² and psychological effects,¹¹³ which may further explain their potential clinical efficacy.

Joint techniques: mobilisation and manipulation

There are a number of different philosophical approaches towards manual treatment of joints, such as those of Cyriax, Kaltenborn, Maitland, McKenzie and Mulligan.¹¹⁴⁻¹¹⁷ Each philosophy differs in its approach to patient assessment and in their justification for treatment selection. The Cyriax and Kaltenborn approaches, for example, base their examination and treatment selection on arthrokinematic and biomechanical theories such as capsular patterns, coupled motions and concave-convex rules. In contrast, the Maitland, McKenzie and Mulligan approaches place more emphasis on patient response during assessment and treatment (symptom provocation and resolution) rather than rigid biomechanical theories.¹¹⁸

Irrespective of the clinical reasoning paradigm, the most common manual techniques applied to joints include mobilisations and manipulations. Mobilisations are rhythmic, low-velocity, oscillatory movement techniques applied to spinal or peripheral joints. They are usually passive techniques performed within the patient's tolerance (Fig. 17.9) and may involve physiological joint motion (movements that patients could perform themselves) and/or accessory motions (movements that patients cannot actively control, such as distraction, compression, sliding, spinning and rolling).

The aim of mobilisation is often to restore full, pain-free range of motion to a joint found to be stiff or painful on clinical examination. Treatment parameters such as the location, direction and magnitude of applied force, and the duration of loading, will vary based on patient

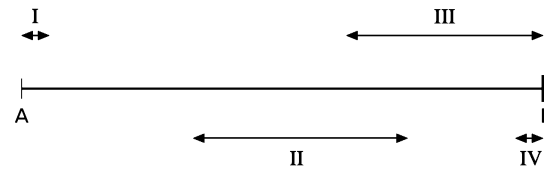


Figure 17.9 Grades of mobilisation: (a) beginning of range of movement; (b) end of range of movement

presentation, clinical examination findings and response to treatment. In general, however, highly irritable joints should initially be treated with low-grade mobilisations (grades I-II, see Table 17.3), whereas higher grades (III-IV) can be used for stiff, non-irritable joints.

Mobilisations may also be combined with active movements (mobilisation with movement, see Fig. 17.10), and in some cases patients may be able to perform mobilisations themselves (self-mobilisation, see Fig. 17.11).

Joint manipulations are high-velocity thrust techniques exerted at the end of joint range, using either a long or a short lever arm. Long-lever techniques move many joints simultaneously, such as rotatory manipulations of the cervical spine. Short-lever techniques involve a low-amplitude thrust aimed at a specific joint or spinal segment. This is achieved through careful positioning of the patient, using arthrokinematic principles to isolate movement in the target joint and 'lock' surrounding joints. Manipulation differs from mobilisation due to the velocity of the technique; patients can theoretically prevent the movement during mobilisation but with manipulation they cannot.

Manipulations are usually associated with an audible 'pop' sound, which is thought to be caused by an event called cavitation (Fig 17.12). This refers to the formation of bubbles (or cavities) within synovial fluid due to reductions in hydrostatic joint pressure. Cavitation is traditionally viewed as the sign of a successful manipulation.¹¹⁹ However, a number of studies have shown that there is little association between an audible pop and the clinical outcomes of spinal manipulation techniques.¹²⁰⁻¹²³

Safety of manipulation techniques

Manipulation has been associated with a variety of adverse effects, ranging from minor short-term reactions

Table 17.3 Maitland's grades of mobilisation¹¹⁷

Grade	Degree of mobilisation
I	Small amplitude movement performed at the beginning of range
II	Large amplitude movement performed within the free range but not moving into any resistance or stiffness
III	Large amplitude movement performed up to the limit of range
IV	Small amplitude movement performed at the limit of range

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Figure 17.10 Mobilisation with movement technique for ankle dorsiflexion

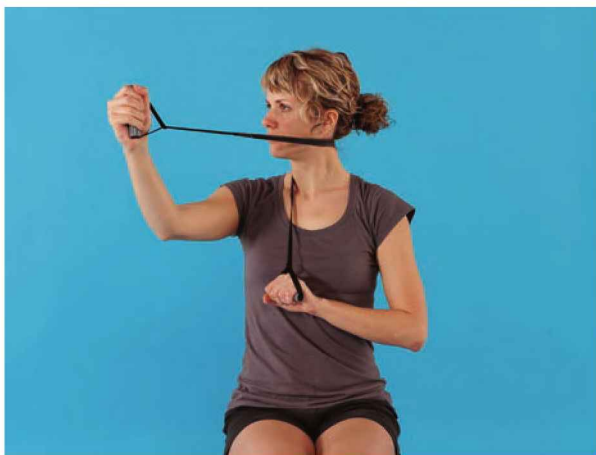


Figure 17.11 Self-mobilisation of cervical rotation

such as headache, stiffness and local discomfort, to severe catastrophic events including stroke and death. Minor reactions can be surprisingly common, with one study finding that more than 60% of patients experienced a mild adverse reaction in the 24 hours after receiving a spinal manipulation.¹²⁴ While a vast majority of side effects are benign, self-limiting problems, clinicians should inform patients of their likelihood prior to treatment.

Fortunately, severe injuries caused by spinal manipulations are extremely rare. While the actual risk is unknown, estimates range from one case in fifty thousand manipulations to one in five million.¹²⁴ A majority of catastrophic injuries caused by cervical spine manipulation are due to the proximity of the vertebral artery to the lateral vertebral joints. However, thoracic and lumbar spine manipulations may also cause serious injury, such as rib fractures and cauda-equina syndrome.¹²⁵⁻¹²⁸

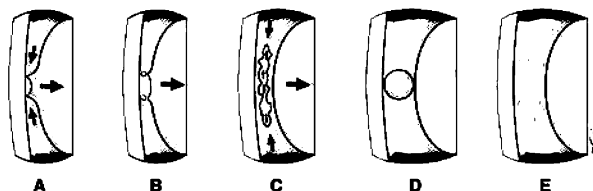


Figure 17.12 The principle of cavitation explains the audible popping sound made during joint manipulation (a) During separation of the joint surfaces, the outer regions of the circular contact zone become pointed. This deformation occurs because at high speeds, the central region of the contact zone separates before the outer region moves, creating a circular rim (b) Joint surfaces snap back at the circular rim where the cavity initially forms, forming small bubbles (c) Small bubbles collect to form a single large bubble (d) The newly formed spherical bubble reaches its maximum size (e) Because of its instability, the single bubble collapses to form a 'cloud' of many smaller bubbles (demonstrable by radiography as a radiolucent region), which later shrink as the gas and vapour dissolve. IMAGE ADAPTED, FROM EVANS.¹¹⁹

Manipulations should only be performed by practitioners with formal training in manipulative skills. Cervical artery function should be tested prior to manipulation of the cervical spine (see Chapter 23), and short-lever techniques should be used in preference to long-lever techniques.

Clinicians also need to be aware of the contraindications to the use of mobilisation and manipulation techniques (Table 17.4).¹¹⁸

Soft tissue therapy

with ROBERT GRANTER

Direct treatment of soft tissues, using hands or treatment tools, to treat specific pain or injury is commonly referred to as soft tissue (or massage) therapy (see Chapter 13 for the use of soft tissue massage techniques in post-exercise recovery). As with other manual treatments, soft tissue therapy is traditionally directed at subjective clinical examination findings such as reduced quantity and quality of joint movement, muscles with increased tone, soft tissue thickening, or muscles containing taut, sensitive bands that may refer pain in distinct patterns, often referred to as myofascial trigger points (Chapter 4).

This section provides an overview of several key techniques that clinicians can use to treat muscular pain, and alert the reader to some self-treatment options.

Treatment position

For successful soft tissue therapy, the clinician should place the target tissue in a position of either tension or laxity.

Table 17.4 Absolute and relative contraindications for mobilisation and manipulation techniques

Absolute contraindications—high risk of a deleterious consequence		
	Mobilisation	Manipulation
Malignancy or tumour in the targeted region	×	×
Cauda equina lesion producing bladder or bowel disturbance	×	×
Fractures	×	×
Local bony infection (e.g. osteomyelitis, tuberculosis)	×	×
Systemic disturbance	×	×
Rheumatoid collagen necrosis	×	×
Unstable upper cervical spine	×	×
Signs of coronary or vertebral artery dysfunction	×	×
Practitioner lack of ability		×
Spondylolisthesis (symptomatic)		×
Gross foraminal encroachment		×
Children/adolescents with open epiphyseal plates		×
Pregnancy (last trimester)		×
Joint fusions		×
Psychogenic disorders		×
Immediately postpartum		×
Relative contraindications—possibility of deleterious consequences		
	Mobilisation	Manipulation
Active inflammatory conditions	×	×
Significant segmental stiffness	×	×
Systemic diseases	×	×
Neurological deterioration	×	×
Irritability	×	×
Osteoporosis	×	×
Condition worsening with present treatment	×	×
Acute radiculopathy	×	×
History and examination do not add up	×	×
Use of oral contraceptives (cervical spine)	×	×
Long-term corticosteroid use (cervical spine)	×	×
Blood clotting disorder	×	×

Treating soft tissue in an elongated position can make focal sites of abnormality more easily palpable, and may enhance referral patterns from myofascial trigger points. Treating in a position of laxity enhances the ability to access and assess deeper layers of soft tissue.

Digital ischaemic pressure

Digital ischaemic pressure describes the application of direct pressure perpendicular to the skin towards the centre of a muscle with sufficient pressure to evoke a temporary ischaemic reaction (Fig. 17.13).



(a)

Figure 17.13 (a) Digital ischaemic pressure to the infraspinatus trigger point using the clinician's thumb
(b) Digital ischaemic pressure to the gluteus medius trigger point using the clinician's elbow



(b)

The aims of this technique are to stimulate the mechanoreceptors within muscle and fascia to reduce resting muscle tone, to provide an analgesic response in soft tissue by eliciting a release of pain-mediating substances, and to deactivate symptomatic trigger points which may optimise muscle activation patterns.¹²⁹

The clinician can apply digital ischaemic pressure using the thumb (Fig. 17.13a), elbow (Fig. 17.13b) or a hand-held device such as a T-shaped bar.

Sustained myofascial tension

Sustained myofascial tension is performed by applying a tensile force with the thumb, braced digits, or forearm (Fig. 17.14) to an assessed site of fascial thickening or reduced fascial glide in the direction of greatest restriction, or in the direction of elongation necessary for normal function. Tension is developed in the tissue by application of compression to the appropriate fascial layer, then moving the thumb, braced digit or forearm through the target tissue to impart a shear force. Greater shear force can be imparted using passive through-range movement or active movement in the direction

of assessed restriction in conjunction with the local tissue contact.

Friction

Friction massage is applied selectively to a localised area palpated as an abnormal tissue thickening; for example, due to excessive intramuscular scar formation following acute muscle injury. It is essential to assess pain and restrictions in muscle length and/or neural mobility before, during and at the end of each treatment session to monitor its effectiveness.

Sufficient pressure must be applied so that the clinician's finger/thumb and the patient's skin move as one, and the friction must work deeply enough and be of sufficient sweep to impart a friction movement to the target tissue. The friction is normally applied at 90° (perpendicular) or at 45° to the target structure, and must be applied at a level that maintains contact with the target tissue and is within the patient's pain tolerance.

It is vital to ensure that the region being treated is totally relaxed. Pressure should be increased gradually to reach the required treatment level within the

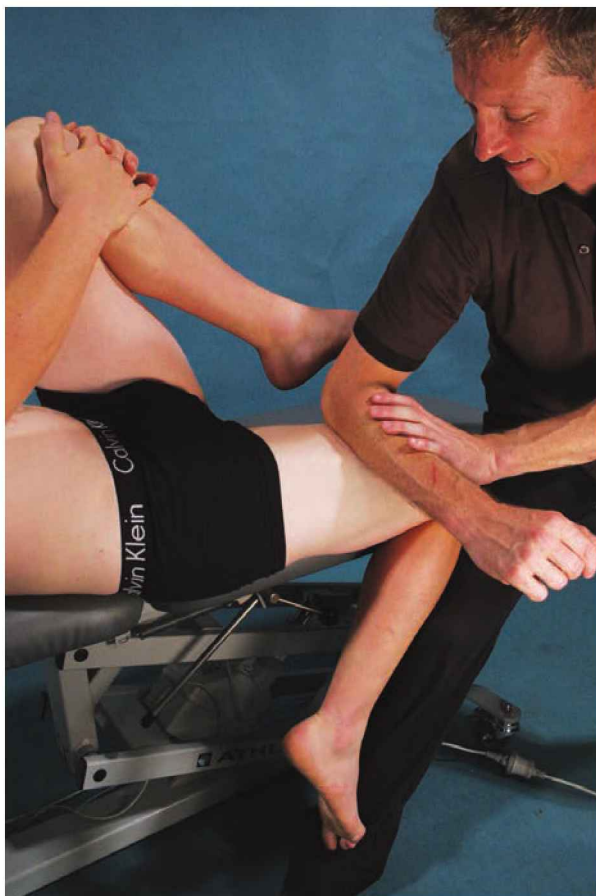


Figure 17.14 Sustained myofascial tension

patient's pain tolerance. Ice can be applied to the region post-treatment in a position of pain-free stretch for 5–10 minutes, particularly for the initial application, to minimise post-treatment tenderness.

Depth of treatment

Granter and King developed a grading system for depth of soft tissue therapy as a guide for clinicians and researchers to adopt consistency in describing treatment dosage (Table 17.5).

The scale of treatment depth is based on two factors: first, the patient's perceived level of pain during the treatment (grade I–IV); and second, the clinician's sense of soft tissue resistance to palpation (A–C).

Combination treatment

If soft tissue techniques are aimed at restoring joint range of motion, they can be followed by static, contract/relax or dynamic stretching to maximise the effect.

Table 17.5 Granter–King scale for grading the depth of soft tissue therapy

Pain grade (P)	Patient's perception of pain
I	No pain perceived
II	Commencement of pain
III	Moderate level of pain
IV	Severe level of pain (seldom used)
Resistance grade (R)	Clinician's perception of tissue resistance
A	No sense of base tissue resistance
B	Onset of base tissue resistance
C	Moderate level of base tissue resistance

Lubricants

Many soft tissue techniques require a lubricant applied to the skin to aid both patient comfort and the clinician's ability to palpate the tissue for abnormalities. There should be sufficient lubricant particularly when palpating areas with large amounts of hair, as irritation of hair follicles may result in contact dermatitis.

With techniques such as sustained myofascial tension, skin contact should be maximal. Therefore, no lubricant is required. As there are fewer repetitive movements used in this technique, there is less risk of irritation to hair follicles.

Vacuum cupping

The aim of vacuum cupping is to restore optimal length and mobility of soft tissues, with changes most likely achieved through neurophysiological responses. Oil is applied to the skin to contain the negative pressure created in the cup by a vacuum pump. Within the vacuum cup, the soft tissue is 'drawn' upwards in a unique perpendicular direction, thereby stretching the soft tissue in a regulated, sustained stretch. The cup contains a one-way valve that allows the pump to be removed.

Cupping can cause significant capillary rupture and damage to the periosteum if used with excessive vacuum pressure or with incorrect placement. The skin colour should be monitored closely and the cup removed if the skin becomes more deeply rose-coloured than normal reactive hyperaemia. Generally, the longer a cup is left *in situ* or the more vacuum force is applied, the more bruising can develop. This is unnecessary and can be avoided by diligent observation during application. The cup should not be applied to vulnerable regions of the body, such as cubital and popliteal fossae and soft tissue attachment sites to bone.

Self-treatment

In recent years, there has been an increase in the popularity of using devices such as spiky balls, tennis balls and foam rollers for self-treatment of soft tissues (Fig. 17.15). A number of studies have found that this approach leads to a short-term increase in range of motion, without any detriment in sports performance.¹³⁰⁻¹³³ Self-treatment may also be aimed at treating myofascial trigger points.

Self-treatment can be undertaken daily and should not cause excessive or lasting pain that adversely affects training or increases symptoms. Typically, self-treatment involves the application of a sustained force to the identified sites, with a sustained pressure until tone/tension reduces and pain or referred symptoms resolve. Functional re-assessment should be carried out after an initial trial period, for example 1–2 minutes, to ensure that positive changes are occurring. As initial presenting symptoms, soft tissue characteristics and patient responses vary, clinicians and patients need to individualise dosage in relation to time and repetitions to achieve optimal results.

Dry needling

The terms ‘acupuncture’ and ‘dry needling’ are often used interchangeably in sports medicine to describe treatments using thin monofilament needles without injectate.¹³⁴ However, *acupuncture* denotes the use of the modality

within a traditional Chinese medical paradigm, in which needles are used to unblock the flow of vital energy, ‘qi’, along meridians or channels,¹³⁵ whereas *dry needling* implies the use of a modern biomedical paradigm.

Dry needling is thought to have a range of neural and biochemical effects that can contribute to pain relief.¹³⁵ Insertion of needles stimulates free nerve endings, sensory receptors and autonomic fibres in skin, muscle and connective tissue. It also causes the release of a range of neuroactive mediators, including endogenous opioids, which can inhibit transmission of nociceptive signals in the peripheral and central nervous system and alter pain perception.^{135, 136} Local tissue injury caused by needle insertion is also thought to cause ‘positive’ inflammatory and immune responses, which can contribute to analgesia.^{134, 135}

Dry needling is most commonly used to treat intramuscular trigger points, but techniques targeting joints, tendons, ligaments, bone and scar tissue are also described.¹³⁴ Trigger point techniques often aim to produce a local muscular twitch response, thought to be caused by stimulation of a spinal reflex loop.¹³⁷ This has been linked to an immediate reduction in extracellular neurotransmitter concentration in muscle tissue surrounding a trigger point,¹³⁸ and a twitch response is anecdotally associated with an improved treatment effect.¹³⁹

There is a vast amount of research on the effects of dry needling and acupuncture. However, interpretation of the literature is complicated by the wide variety of conditions treated, differing underlying treatment paradigms (traditional Chinese medicine versus a modern biomedical model), as well as a large variation in treatment protocols and methodological quality. It is particularly difficult to blind patients and clinicians in RCTs which is problematic as needling is associated with a potent placebo effect.

Nevertheless, a substantial body of level 1 evidence suggests needling is more effective than placebo or sham treatments for a range of musculoskeletal conditions such as chronic lower back, neck and shoulder pain, chronic muscular pain, headache, and knee osteoarthritis.¹⁴⁰⁻¹⁴⁴

Clinicians should consider the following points when using dry-needling techniques.

- The patient is comfortably positioned with the target region well-supported.
- The target tissue is clearly identified, a taut band is palpated within the muscle and symptoms are reproduced by palpation. This site is ‘marked’ by compressing a used guide-tube onto the skin above the identified site (Fig. 17.16a).
- The correct equipment is prepared. Sterile, single-use needles must be used. Tubed needles are recommended



Figure 17.15 Self-treatment: treating the gluteus medius with a tennis ball

to expedite placement and reduce the risk of touching the needle shaft. Cotton swabs and a sharps container should be within easy reach.

- A hygiene protocol is followed. The patient's skin is cleaned and disinfected with an alcohol swab. The clinician should wash his or her hands and wear latex gloves (at least on the palpating hand most likely to be exposed to blood).



(a)



(b)

Figure 17.16 Dry needling (a) A needle tube can be pressed into the skin to temporarily mark the injection site (b) The target muscle should be supported by one hand while the other hand focuses on accurate insertion of the needle

- The target muscle is supported to maximise accurate needle penetration. For example, when treating the upper trapezius, the muscle should be held anteriorly and posteriorly between the fingers and thumbs of one hand while the other hand focuses on accurate insertion of the needle (Fig. 17.16b).
- The needle and guide tube are placed perpendicular to the skin on the marked site and the needle is inserted with a short, sharp tap to the top of its handle.
- The guide tube is removed and the needle is fully inserted to the appropriate depth by only touching the needle handle, never the shaft.
- Various dry needling techniques can be utilised (see below).
- The needle is removed and local compression is applied over the site via a cotton swab for approximately 30 seconds.
- The joints associated with the muscle treated should be moved passively through their full comfortable range of motion after treatment.
- Needles and equipment are disposed of by correct medical waste disposal protocols.

The most commonly used dry needling technique involves inserting the needle into focal trigger points within a palpable taut band and leaving it in place for a few minutes. There is no clear consensus on the time left in situ—some authors have suggested time ranges from 10–30 minutes.¹³⁴ This will vary according to patients' experience with dry needling, as well as their emotional and physiological responses and the degree of presenting symptoms.

Some clinicians also apply electrical stimulation to the inserted needles. However, there are no clear guidelines for treatment parameters and no evidence that electrostimulation improves the treatment effect.

Alternatively, the clinician can insert the needle into the muscle and then repeatedly move the needle in and out (such that it still remains in the sub-dermal region), constantly trying to find specific points that reproduce the patient's local or referred pain, or produce a 'twitch' response. Initially, the needle may be 'grasped' by the muscle, followed by a gradual relaxation.

Clinicians can apply needles to a number of trigger points during each treatment session. Usually, pain relief lasts 3–4 days after the first treatment session. The duration of the pain relief may be longer following subsequent sessions. Up to three or four treatments may be required initially to eliminate a trigger point, and no single trigger point should be needled more than twice in a week.

Various research studies have compared dry needling to injection of local anaesthetic and various other substances. There is some evidence that dry needling can be just as effective.^{145, 146}

Risks of adverse events during dry needling

The most common adverse effect associated with dry needling is soreness in the first 24 hours post treatment.¹⁴⁷ Application of heat and stretching exercises may minimise this,¹⁴⁷ and aggressive treatment should be avoided until the patient's reaction to dry needling is known. High quality needles, which are sharper and thinner, may also reduce pain during and after treatment. If there is excessive resistance to needle removal, the needle should be removed more slowly.

In patients unfamiliar with or apprehensive about dry needling, it is possible to induce an episode of vasovagal syncope. The risk of this occurring can be minimised by careful patient selection—it is especially important to identify those with needle phobias. Clinicians should avoid excessive stimulation of specific sites until the patient learns to tolerate this new treatment. The patient's response should be closely monitored during treatment, with constant communication to ensure patient comfort. If at any stage the patient becomes distressed with the treatment, needles should be removed. Contraindications to dry needling include bleeding disorders, active infection, blood-borne diseases, allergies to metal, unstable epilepsy and the third trimester of pregnancy. Additionally, extreme care should be taken when dry needling near the lungs as there is a risk of causing a pneumothorax.

Neurodynamic techniques

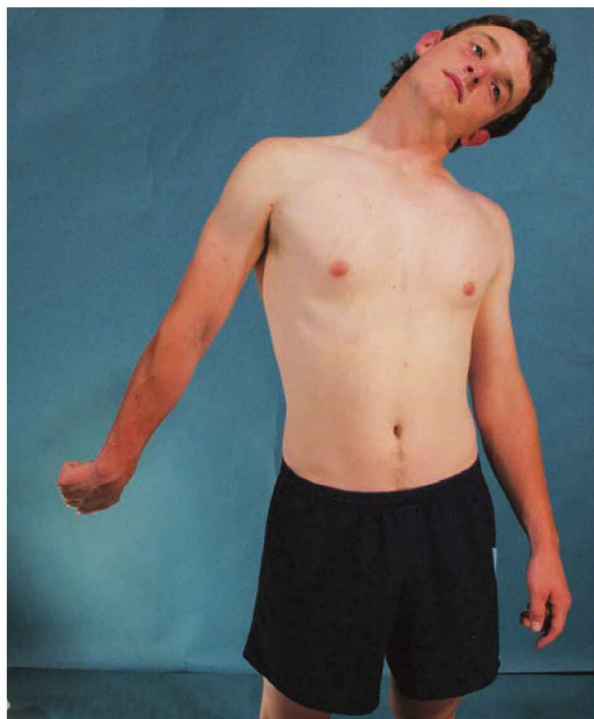
The nervous system needs to adapt to mechanical loads. It can do this through elongation, sliding, cross-section change, angulation and compression.¹⁴⁸ However, when the nervous system cannot cope under loads, it can lead to neural oedema, ischaemia, fibrosis and hypoxia—all of which can alter neurodynamics.¹⁴⁸ This change in neurodynamics may make a significant contribution to the patient's symptoms and signs in certain injuries. Unless these abnormalities are addressed in addition to other soft tissue abnormalities associated with the injury, full recovery as indicated by full pain-free range of motion may not occur.

Neural tension was a term used in the past to describe peripheral nervous system dysfunction, but was thought to only address the mechanical aspects of the dysfunction.¹⁴⁸ Nowadays the term *neurodynamics* is employed to describe the biomechanical, physiological and structural dysfunction of the nervous system.¹⁴⁸

Neural stretching (mobilisation) is a treatment modality that addresses neurodynamics. Neural stretches aim to restore dynamic balance of the nervous system.¹⁴⁸ Proposed benefits include facilitation of nerve gliding, decrease in nerve adherence, dispersion of noxious fluids, increased neural vascularity and improved axoplasmic flow.¹⁴⁸

These stretches are adaptations of the neural tension tests (Chapter 15). The two most commonly used neural stretches are adaptations of the upper limb tension test (Fig. 17.17a) and the slump test (Fig. 17.17b).

A systematic review¹⁴⁸ evaluated the effectiveness of neural mobilisations and found that even though the identified studies did report positive effects, low methodological quality and limited number of studies suggest that currently there is limited evidence for their use. However, these stretches can often be helpful in the treatment of conditions in which neurodynamics are



(a)



(b)

Figure 17.17 Common neurodynamic techniques (a) Upper limb neurodynamics stretch test (b) Slump stretch

abnormal, and variations of these tests may be used for both diagnosis and treatment.

Clinicians should take particular care when considering neural mobilisation exercises for acute or irritable conditions, as they may aggravate the patient's symptoms. Stretches should always begin gently, and gradually increase under the close supervision of an experienced clinician. As with other methods of treatment, neural stretches alone are rarely sufficient to correct all abnormalities present. They can be particularly effective in long-standing, chronic conditions where neural hypersensitivity is common.

TAPING

with SAM BLANCHARD

Taping was the third-most utilised intervention at the London 2012 Olympic Games polyclinic,¹⁴⁹ reflecting its current popularity as a treatment modality and injury prevention intervention. However, research into both the mechanisms and effects of taping is limited and marked by heterogeneous results,^{150, 151} making it difficult for clinicians to draw firm evidence-based conclusions on its benefits and application.

Proposed mechanisms of taping

Taping is proposed to have a range of effects depending on the application method and the type of tape used. Traditionally, clinicians have applied rigid strapping tape in an attempt to restrict patients' range of motion or to alter their anatomical alignment; for example, to realign the patella within the femoral trochlea (Fig. 17.18).¹⁵² More recently, the use of flexible cloth tape, often referred to as kinesiology tape, has also gained popularity.

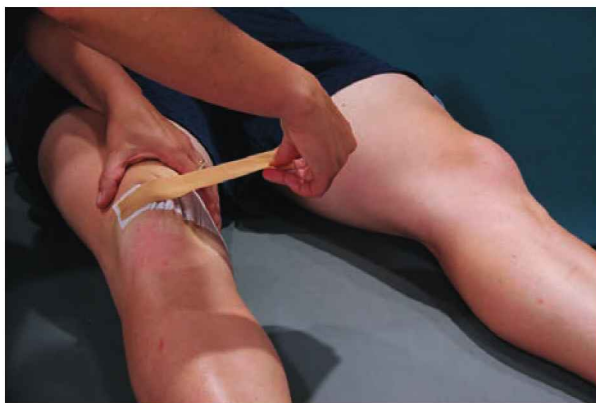


Figure 17.18 Taping is often used to relieve patellofemoral pain

Taping also appears to have a range of neurophysiological effects.

- Cutaneous stimulation by tape may inhibit afferent nociceptive signals at the spinal level (pain-gate theory).¹⁵³
- Tape application has been shown to alter brain activity in a variety of areas, including the motor cortex, sensory cortex, thalamus, basal ganglia and cerebellum.¹⁵⁴ This may have a wide range of effects, including improved proprioception¹⁵⁵ and motor coordination, and explain how taping influences factors such as muscle activation timing¹⁵⁶ and concentric/eccentric control.¹⁵⁷
- Restriction of the range and amplitude of joint motion may enable increased reaction time of surrounding musculature during high-velocity injury situations.^{150, 157-159}
- Patients' conscious and subconscious expectations of the effects of taping may relieve pain (i.e. placebo effect).¹⁵³ Once thought to be a 'purely psychological' phenomenon, it is now considered to be at least partly caused by the secretion of endogenous opioids in response to treatment.⁵⁸

However, the proposed mechanisms of taping remain largely speculative or controversial. For example, tape intended to limit range of motion has been shown to loosen after 10-30 minutes,^{160, 161} and recent systematic reviews disagree on whether patellofemoral joint taping influences VMO onset timing and knee extensor moments.^{156, 162} Research in this area is ongoing.

Evidence of efficacy

Although taping is used for a wide range of conditions across the body, high-level evidence of its efficacy is scant. The available evidence is of varying quality—one of the main challenges when studying taping is the inability to blind patients or practitioners.^{156, 166}

Patellofemoral pain

A number of systematic reviews have investigated the use of patellar realignment taping (often referred to as McConnell taping—see Chapter 36) in patients with patellofemoral pain syndrome.^{156, 166, 167} Although their results are somewhat conflicting, there is moderate evidence that taping provides an immediate reduction in pain,^{156, 167} particularly when an individualised taping technique is used.¹⁵⁶ However, its benefit seems to be limited to the short term, and prolonged use is not supported. The effect of kinesiology tape on patellofemoral pain remains unclear; one randomised crossover trial found that McConnell and kinesiology taping were equally effective in immediately reducing pain,¹⁶⁸ whereas two

Kinesiology tape: valuable tool for athletes or fashion statement?

Kinesiology tape has gained considerable popularity in recent years, as demonstrated by the large number of athletes using it at the Olympic Games and other major sporting events (Fig. 17.19). The approach was developed in Japan in the 1970s and is claimed to permit unrestricted range of movement while supporting the fascia, muscles and joints.¹⁶³ Kinesiology tape has been proposed to improve pain and blood flow via dermal lifting, which decreases the pressure on cutaneous nerve receptors and capillaries, decreasing nociceptive input and increasing afferent feedback.¹⁶³ However, these effects last less than 24 hours and may be trivial, and the actual benefits for injured athletes are yet to be established in high-quality studies.¹⁶⁴ The use of kinesiology tape remains controversial among clinicians, with some even claiming it may provide a psychological crutch for athletes.¹⁶⁵



Figure 17.19 Kinesiology tape has gained popularity among athletes in recent years (note player on left, tape on knee and thigh) IMAGE COURTESY OF ROCKTAPE UK

randomised controlled trials found kinesiology tape was ineffective in reducing pain.^{169, 170}

Ankle sprains

Three studies have suggested that taping is effective in prevention of ankle sprains.¹⁷¹⁻¹⁷³ However, the highest-quality prevention research has focussed on the effects of rigid or semi-rigid prefabricated braces, and it is unclear whether taping is equally effective.¹⁷⁴ Irrespective of the form of support used, any preventive effect of taping and bracing seems to be limited to athletes with a history of ankle sprain. As the risk of re-injury is increased sevenfold in the year following an initial sprain,¹⁷⁴ taping or bracing could be used in that time (Chapter 12).

Other conditions

There is some evidence that taping may be effective in treating pain and strength with lateral epicondylalgia,¹⁷⁵ de Quervain's tenosynovitis,¹⁷⁶ plantar fasciopathy pain^{153, 177, 178} and shoulder pain.^{153, 163, 179} However, many of these studies are of low quality with a moderate to high risk of publication bias. These conditions are covered in Part B.

Practical considerations

In practice, taping is applied across a far wider range of conditions than there exists evidence for, and should be considered just as much an art as a science.¹⁷⁴ Clinicians should consider not only the available literature, but also the diagnosis, the scenario (e.g. competition, acute management, rehabilitation), and the patient's preference and expectations.

We recommend the use of a trial-and-error approach. A 'treatment direction test', first proposed by Professor Bill Vicenzino for the prescription of orthotic insoles,¹⁸⁰ can be applied whenever tape is used to relieve pain. In order for the intervention to be judged worthwhile, it should lead to a substantial reduction in symptoms (at least 50% on a visual analogue scale) during a relevant functional task.

Much like any exercise program should be progressive, the use of taping should be gradually reduced over time.⁷² When taping is successful in alleviating a chronic condition such as patellofemoral pain, clinicians should make a plan to wean the patient off the use of tape to avoid creating a dependence on a passive treatment. Following an acute injury, taping may provide immediate pain relief and restrict excessive movement.¹⁸¹ Controlled stress (optimal loading) can then be facilitated by gradually reducing the amount of support taping provides over subsequent days and weeks.

PRACTICE PEARL

For long-term application of tape, consider teaching the athlete to tape him- or herself. This will increase the sense of control over his or her own management, decrease dependency on the clinician and save valuable time in a busy treatment clinic.

In order to protect the patient's skin from irritation, hypoallergenic tape can be applied under rigid sports tape. This is particularly important when tape containing zinc

oxide is used for longer durations. In some situations, such as prophylactic ankle taping, pre-wrap may be applied under the tape. This does not alter the amount of motion restriction, the time to maximum range or the velocity of range;¹⁵⁹ however, it may reduce the cutaneous stimulus.

PRACTICE PEARL

Where possible, new taping techniques should be introduced to athletes in a controlled environment prior to competition.

Despite the limited and conflicting evidence base, taping continues to play an important role in clinical sports medicine practice. Clinicians should consider the available literature, retain a healthy scepticism towards commercial claims and trends, and take a trial-and-error approach. Clinical experience and athlete preference should not be discredited.

ELECTROPHYSICAL AGENTS

with NICK GARDINER, ADAM GLEDHILL, LAWRENCE MAYHEW and VASILEIOS KORAKAKIS

Electrophysical agents (electrotherapy) have been used to treat musculoskeletal conditions for over 70 years.¹⁸² Although their popularity appears to be waning in favour of active treatments, traditional modalities such as ultrasound and interferential stimulation, and newer ones such as extracorporeal shockwave therapy remain highly used in clinical practice. In this section, we review the proposed mechanisms and evidence of effect of the most common electrophysical agents currently used in sports medicine.

The decision of whether to use electrotherapy will be individual and influenced by factors such as time and resource availability, patient expectation and clinician preference. However, it should be stressed that electrophysical agents should never be used in isolation. If they are to be used at all, it should be as an adjunct to other forms of treatment, advice and education.

Therapeutic ultrasound

Therapeutic ultrasound has traditionally played a large role in physiotherapy practice around the world,¹⁸³⁻¹⁸⁸ and remains a popular treatment modality among physiotherapists and other health professionals such as osteopaths, chiropractors and sports therapists.^{183, 189}

A range of proposed effects of ultrasound form the biological rationale for its therapeutic use. These include thermal effects such as increased local blood flow, metabolic activity and collagen extensibility,

as well as non-thermal effects such as stimulation of fibroblast activity, mast cell degranulation, growth factor production and angiogenesis.¹⁹⁰ Although these effects have largely only been demonstrated *in vitro*,^{183, 190} they are commonly cited to justify the clinical use of ultrasound to treat a wide variety of musculoskeletal conditions, with the belief that it can reduce pain, inflammation and oedema, and enhance scar tissue remodelling, tissue extensibility and soft tissue healing.¹⁸⁹ Treatment is traditionally directed towards collagen-rich tissues such as ligament, tendon, fascia, joint capsule and scar tissue, as these are thought to absorb the greatest amount of energy.¹⁸³

Despite the theoretical appeal of ultrasound, however, there is little high-quality evidence to support its clinical effectiveness.^{182, 191, 192} For example, a recent systematic review of the effect of ultrasound on a range of lower-limb soft tissue conditions found that no placebo-controlled trials were able to demonstrate a significant difference between true and sham ultrasound therapy.¹⁹¹

Although some argue that this may be due to heterogeneous study populations, conditions treated and dosage parameters,¹⁸³ the current lack of research evidence supporting therapeutic ultrasound for soft tissue conditions should be taken into account by clinicians considering its use.

Ultrasound as a stimulator of bone repair

In recent years, ultrasound has been increasingly used in the treatment of acute bone fractures. Although this was traditionally thought to be contraindicated, recent meta-analyses suggest that daily administration of low-intensity pulsed ultrasound (LIPUS) can accelerate the healing of fractures by approximately 35%.¹⁹³⁻¹⁹⁵

LIPUS appears to be most effective for conservatively managed diaphyseal fractures, reducing the time to clinical union by approximately 18 days.¹⁹⁵ The evidence is strongest for upper limb fractures, whereas it is conflicting for metaphyseal and operatively managed acute fractures.^{193, 195}

LIPUS is typically applied for 20 minutes every day, which may be an excessive treatment regimen for many patients. However, in elite sports environments a faster return to training and competition might be very important. LIPUS may also be valuable in cases that fail conservative management (non-union) after bone-graft surgery is performed.¹⁹³

Few studies have investigated LIPUS for the treatment of stress fractures; however, the available results are not encouraging^{196, 197} and its use cannot currently be recommended for this purpose. Similarly, there is currently no evidence that LIPUS is effective for soft tissue injuries.

Transcutaneous electrical nerve stimulation

Transcutaneous electrical nerve stimulation (TENS) delivers biphasic electrical current with the use of electrodes applied to the skin.¹⁹⁸ There are several mechanisms by which TENS can provide short-term pain relief, including stimulation of opioid and noradrenergic receptors at the site of application, and opioid, serotonin and muscarinic receptors in the spinal cord and brainstem.¹⁹⁹

Although there is no evidence that TENS improves long-term outcomes,²⁰⁰ it may play an adjunct role in pain management in certain patient groups. It is reported to provide effective short-term pain relief in patients with spinal cord injuries,^{201, 202} lumbar disc disease,²⁰³ and chronic neck and lower back pain.²⁰⁴ There is currently little evidence to support its use for knee osteoarthritis^{198, 205} and it is not widely used in the athletic population.

TENS can be used in combination with exercise and medication.^{201, 206} Potential adverse effects include mild erythema and itching underneath the electrodes.²⁰⁶

Neuromuscular stimulators

Neuromuscular electrical stimulation (NMES) is purported to have a range of effects on healthy athletes, including enhanced recovery, improved sports performance and reduced muscle soreness. However, as discussed in Chapter 13, commercial claims regarding the benefits of NMES for athletes often go well beyond the existing evidence.²⁰⁷ In a clinical setting, NMES may be valuable to enhance muscle strengthening after a major injury such as anterior cruciate ligament rupture.²⁰⁸

Traditionally, it has been thought that NMES improves muscle performance because it reverses the normal motor unit recruitment order, preferentially activating larger motor units composed primarily of fast-twitch fibres.²⁰⁹ However, this theory has recently been challenged, with evidence suggesting that the normal clinical methods for applying NMES cause random, non-selective motor unit activation.²¹⁰ Nevertheless, there is high-level evidence

that combining NMES with traditional exercise training is more effective than training alone in improving quadriceps strength following anterior cruciate ligament reconstruction.²⁰⁸

The decision of whether or not to use NMES in rehabilitation is dependent on a number of factors. Effective therapeutic intensity may be uncomfortable, so a high degree of patient motivation is necessary. Clinicians should also remember that, for many patients, traditional strength training produces the desired results. However, a subset of patients experience persistent weakness following major joint injury due to arthrogenic muscle inhibition.²¹¹ In this group, the use of NMES in combination with heavy strength training may be recommended.²¹²

The recommended parameters to enhance muscle strengthening in rehabilitation are shown in Table 17.6.

Interferential stimulation

Like therapeutic ultrasound, interferential stimulation has long played a role in traditional physiotherapy practice and remains widely used in some areas of the world.^{215, 216} It is characterised by the interference of two medium-frequency currents passed through the tissues, causing analgesic effects. The analgesic effect may be due to stimulation of cutaneous receptors that inhibit c-fibre nociceptive transmission,²¹⁷ motor stimulation increasing the removal of pain-inducing substances²¹⁸ and the release of endogenous opioids.²¹⁹

The pain-relieving effects of interferential stimulation are transient and there is little evidence to support its use in athletic populations. However, its efficacy in the management of knee osteoarthritis^{198, 217} suggests potential benefits for athletes following major knee surgery such as ACL reconstruction.

Laser

Low-level laser therapy (LLLT) is a non-invasive treatment in which non-thermal laser irradiation is applied to the site of pain using light generated by high-intensity electrical stimulation of a medium. It modulates cell and tissue

Table 17.6 Recommended treatment parameters to maximise muscle strength gains using NMES^{210, 213, 214}

Stimulation intensity	The intensity should be as high as the patient can tolerate, and at least 50% of MVC
Pulse frequency	High-frequency stimulation (50–80 Hz) leads to higher force production, smoother contractions and greater muscle fatigue. Frequencies over 80 Hz may be more painful
Pulse duration (width)	A longer pulse duration (up to 600 µs) maximises force production for a given stimulation intensity. The currently recommended duration is 200–400 µs
Training regimen	3–10 s ‘on-time’ followed by 10–40 s recovery; 10–15 min per session
Program parameters	1–3 sessions per week for 4–6 weeks

physiology to obtain therapeutic effects,²²⁰ including promotion of tissue repair,^{221, 222} and inflammatory effects,^{223, 224} thereby providing pain relief in common musculoskeletal disorders.^{220, 225, 226}

LLLT may assist in pain relief following acute sports injury²²⁷ and has been found to reduce pain and improve function in patients with rotator cuff tendinopathy, both as a stand-alone treatment and when used in combination with therapeutic exercise.²²⁸ It may also improve outcomes for patients with chronic midportion Achilles tendinopathy when combined with an eccentric loading programme (Chapter 40).²²⁹ Although there is some evidence that LLLT is an effective treatment in certain conditions, there is little knowledge of dose-response and no consensus on optimal treatment parameters for specific pathologies. Further research utilising large, homogenous samples of similar pathologies would clarify the efficacy of LLLT in sports injury management.²²⁷

Electromagnetic therapy

Low-frequency pulsed electromagnetic field therapy (PEMF) is purported to have a wide range of local cellular and systemic effects affecting the healing process. Its use is recommended in wound care,^{230, 231} and there is some evidence supporting the use of PEMF in the treatment of certain sports injuries.

Laboratory studies suggest that it may be valuable in the treatment of tendon, bone and cartilage pathology.²³²⁻²³⁵ For example, when applied to human tendon cells, PEMF may cause cellular proliferation, tendon-specific gene expression, the release of growth factors, and the release of pro- and anti-inflammatory cytokines.^{234, 235} Similarly, when applied to articular hyaline cartilage, PEMF can stimulate chondrocyte proliferation and the stimulation of proteoglycans and extra-cellular matrix.²³²

The results of clinical studies of PEMF in athletic populations are mixed.

- In athletes who underwent arthroscopic treatment of osteochondral defects of the talus, PEMF did not lead to earlier return to play than did a placebo treatment.²³⁶
- Compared to a placebo, patients who received PEMF after arthroscopic knee surgery had improved functional outcomes at 45 and 90 days and at a 3-year follow-up.²³⁷
- PEMF may have short-term functional, analgesic and anti-inflammatory benefits in shoulder and elbow tendon conditions.^{238, 239} However, there is little evidence of any long-term benefits.^{239, 240}
- A systematic review of the effects of PEMF and LIPUS on bone injuries found a reduction in healing time following acute diaphyseal fractures.¹⁹⁵ However, due to the small number of studies investigating PEMF, no conclusions on its specific effects could be made.

Extracorporeal shockwave therapy

In 1980, extracorporeal shockwaves were first used in medicine to destroy kidney stones.^{241, 242} Subsequent animal research on extracorporeal shockwave therapy (ECSWT) demonstrated dose-dependent destructive (high-energy) and regenerative (lower-energy) effects on a range of tissues, including bone and tendon.^{243, 244} Since then, ECSWT has been used extensively in clinical practice to treat superficial musculoskeletal conditions (Fig. 17.20).^{245, 246}

Several RCTs and case-control studies have shown ECSWT, either alone or in combination with exercises, to be beneficial for a number of musculoskeletal disorders including plantar fasciopathy,^{247, 248} midportion and insertional Achilles tendinopathy,²⁴⁹⁻²⁵³ medial tibial stress syndrome,^{254, 255} patellar tendinopathy,^{256, 257} proximal hamstring tendinopathy,²⁵⁸ greater trochanteric pain syndrome,²⁵⁹ medial and lateral epicondylitis,^{260, 261} and calcific supraspinatus tendinopathy.^{262, 263}

There is a growing body of evidence supporting the clinical use of shock wave therapy. However, even though level 1 and level 2 studies are available for data pooling and meta-analyses, clinicians should interpret the results of systematic reviews with caution. This is because the included studies are often marked by methodological flaws, biases and low reporting quality.²⁶⁴⁻²⁶⁶ Furthermore, there is inconsistent reporting of the type of shock wave used (extracorporeal, radial, focused), the intensity (high, medium, low) and the method of application (focal, local, area of pain distribution).

Clinicians should consider the following when using ECSWT in clinical practice.

- ECSWT is not a panacea—its efficacy varies considerably for individual pathologies, and careful selection of patients is essential.^{267, 268} For example, it is only effective in treating supraspinatus tendinopathy when



Figure 17.20 Extracorporeal shock wave therapy

calcification is present. For normal (non-calcified) supraspinatus tendinopathy, it has no effect.²⁴⁵

- There is no universal agreement regarding the definition of low-, medium- and high-energy flux density.^{244, 263} Resulting energy is dependent on the device and applicator and should be calculated.
- There is evidence of a dose-related ECSWT effect, with lower-energy flux densities requiring more sessions to obtain the same result.²⁶⁹
- Although there is no consensus on treatment parameters, successful RCTs have used 1–3 sessions of 1500–2000 impulses at weekly intervals,²⁴⁶ regardless of the underlying pathology.²⁴¹
- Treatment parameters should be individualised based on the patient's presentation. For example, the pre-treatment symptom duration was significantly correlated with the number of ECSWT sessions applied.²⁷⁰
- As for all electrophysical agents, ECSWT should not be used in isolation. Prescribing therapeutic exercise in addition to shock waves can lead to better results.^{250, 254}

THERAPEUTIC MEDICATION IN MUSCULOSKELETAL INJURY

with NOEL POLLOCK

Therapeutic drugs can be helpful in the management of musculoskeletal injury by treating pain, modulating inflammation and augmenting recovery. The use of therapeutic drugs should be aligned with the overall management plan and support the particular goals of that management phase. This section will discuss a range of oral and injectable therapeutic drugs used in the management of musculoskeletal injury.

Doctors prescribing medications to athletes should be fully aware of the current anti-doping regulations for the sport in which their patient is participating. This is updated annually by the World Anti-Doping Agency (WADA) and the current list can be found on the WADA website. There is also an online anti-doping resource for athletes and clinicians for products sold in Canada, USA, UK, Japan and Australia accessible at www.globaldro.com where medication can be checked. The site also contains links for other nationalities.

Analgesics

Analgesic medication is used to relieve and modulate pain, and is often prescribed in the acute phase after injury to reduce pain in conjunction with ice, compression or splinting. Analgesia may also facilitate early movement, although the medical team should decide the optimal initial load based on the diagnosis. Analgesic medication may also have a role in other management phases to treat central sensitisation or to facilitate desired movements in rehabilitation.

Paracetamol

Paracetamol (acetaminophen) has an analgesic and antipyretic effect without affecting the inflammatory process. The mechanism of action is probably through the modulation of cannabinoid receptors. Adult oral doses are 1000 mg every 4–6 hours to a maximum daily dose of 4 g. At these doses, paracetamol is well tolerated and can be an excellent choice in acute sports injuries. Soluble paracetamol is more quickly absorbed and may be an effective choice in athletes who need a quicker onset of analgesia. At higher doses, paracetamol is hepatotoxic and it may be fatal in overdose.

Codeine

Codeine is a more potent opiate analgesic and may be considered in the initial treatment of painful acute injury. However, codeine preparations are commonly associated with side effects such as nausea, dizziness and constipation, which limits prolonged use in the treatment of athletes.

Emergency analgesia

In severe acute musculoskeletal injury, stronger and parenteral analgesia may be required. Clinicians that provide event cover should be familiar with the indications and risks of these medications and have access to appropriate emergency analgesia. This may include opiate-based medication, ketamine and Entonox[®].

Non-steroidal anti-inflammatory drugs (NSAIDs)

NSAIDs are drugs with analgesic and anti-inflammatory properties. The term non-steroidal is used to distinguish these drugs from corticosteroids that also produce anti-inflammatory effects. NSAIDs' mechanism of action is to inhibit the cyclo-oxygenase (COX) system. Cyclo-oxygenase has a key role in the inflammatory cascade that occurs at the site of acute injury. It converts arachidonic acid to prostaglandins and thromboxane which are key mediators of inflammation.²⁷¹ There are two cyclooxygenase iso-enzymes, COX-1 and COX-2, with different physiological roles (Fig. 17.21).

COX-1 is associated with the maintenance of the gastric mucosa and COX-1 inhibition with traditional NSAIDs can result in gastritis and ulceration. Selective COX-2 inhibitors have been available since 1999 with similar efficacy, but fewer gastro-intestinal side effects. The most commonly prescribed NSAIDs include aspirin, ibuprofen, diclofenac and naproxen, and the COX-2 selective NSAIDs, celecoxib and etoricoxib. The normally recommended dosage and frequency of administration varies between the different drugs (Table 17.7).

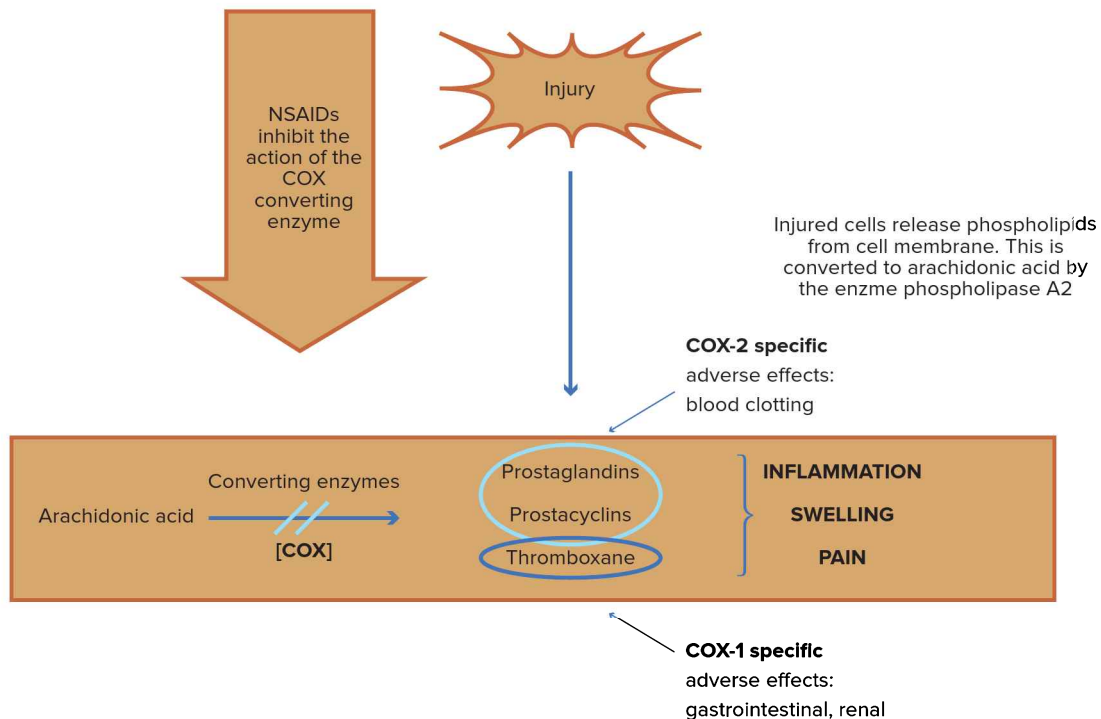


Figure 17.21 Mechanism of action of NSAIDs

Table 17.7 Commonly used NSAIDs

Drug	Some trade names	Usual dose (mg)	Half-life	Daily doses
Acetylsalicylic acid (ASA)	Aspirin®	650	30 mins	3–4
Celecoxib	Celebrex®	100–200	11–12 hours	1–2
Diclofenac	Voltaren®, Voltarol®	25–50	1–2 hours	2–3
Ibuprofen	Brufen®, Motrin®, Advil®	400	1–2.5 hours	3–4
Meloxicam	Mobic®	7.5–15	20–24 hours	1
Naproxen	Naprosyn, Anaprox®	250–1000	12–15 hours	1–2

Topical anti-inflammatory preparations are also available in creams and prolonged-release patches. These are effective delivery methods for superficial injuries, with less risk of side effects than orally administered NSAIDs.²⁷² Overnight patches or wraps may be particularly useful in limiting early morning stiffness that may be associated with excessive inflammation.

There is high-level evidence that topical NSAIDs are effective in relieving pain associated with acute musculoskeletal injuries such as sprain, strains and contusions,^{272, 273} as well as with osteoarthritis.²⁷⁴ Some of the pain relief they provide can be explained by a powerful placebo effect, which is greater with topical NSAID administration than with oral.

NSAID use in sport

NSAIDs are widely used in competitive sport, with more than a third of all athletes at Olympic Games,²⁷⁵ half of all players at recent soccer World Cups²⁷⁶ and about a quarter of athletes in international track and field athletics²⁷⁷ reporting in-competition NSAID use. They are commonly used in the treatment of injury, but also prior to competition in an attempt to reduce discomfort of potential future injury or post-exercise soreness in both elite and non-elite athletes.^{278–280} This practice should be discouraged, as there is no evidence of a reduction in delayed onset muscle soreness (DOMS) or improved performance with prophylactic NSAIDs. However, player perception and the placebo effect can make behaviour

change challenging for team clinicians. Regular NSAID use causes significant health risks for the cardiovascular, renal and gastrointestinal systems. Clinicians should educate their athletes about these risks and seek safer alternatives, perhaps from nutraceutical products.

NSAID use in the treatment of musculoskeletal injury

NSAIDs are usually an excellent choice in the short-term treatment of inflammatory conditions, such as bursitis or synovitis. However, the impact of NSAIDs on cellular metabolism and the tissues of the musculoskeletal system should be considered by athletes and clinicians using these medications in the treatment or prevention of injury. With this knowledge, NSAIDs may be prescribed for particular injuries or in stages of healing determined by the overall goals of the current rehabilitation and management phase.

NSAID and muscle

At a cellular level, NSAIDs probably have a negative effect on both muscle healing following injury and muscle hypertrophy after athletic training.²⁸¹ NSAIDs can inhibit protein synthesis²⁸² and reduce satellite cell activation, which is an essential step in muscle repair and regeneration.^{283, 284} Inhibition of protein synthesis after resistance exercise may be limited to COX-1 NSAIDs, as selective COX-2 inhibitors have not been shown to induce similar reductions in protein synthesis.²⁸⁵

NSAIDs are often used by athletes in the treatment and prevention of DOMS, but scientific trials demonstrate very little evidence for a positive effect.

NSAIDs have been recommended for use in deep muscle contusions at risk of developing myositis ossificans.²⁸⁶ Regular indomethacin for more than seven days is widely used by clinicians in this condition.

NSAIDs are not effective in the treatment of acute muscle tears and cannot be routinely recommended.^{287, 288} They may have a short-term role to modulate an excessive early inflammatory response, but as this phase is necessary for appropriate satellite cell activation and initiation of regeneration, their use at this time may be detrimental to overall muscle repair.

NSAID and tendon

Basic science and animal-model studies of tendon tearing or surgical repair have consistently shown that NSAIDs reduce tenocyte proliferation and collagen formation, and therefore negatively affect tendon healing. However, in clinical studies of acute reactive tendinopathy, NSAIDs may facilitate short-term improvements in pain.²⁸⁹ One small study also demonstrated improvements in leg stiffness and tendon function following treatment with a COX-2 inhibitor.²⁹⁰

It has been suggested that they may be useful in acute tendon overload despite (or perhaps because of) the potential negative effects on tenocyte activation.²⁶⁸ However, clinicians should remember that basic science and clinical studies have shown that NSAIDs blunt exercise-induced collagen synthesis within tendon tissue.²⁹¹ Therefore, the prescribing of NSAIDs should, as with all medication, be considered within the overall treatment goals of the current rehabilitation phase. In Cook and Purdam's model of a continuum of tendon pathology (Chapter 4),²⁶⁸ NSAIDs may be considered in the reactive tendinopathy phase, but are not recommended in the dysrepair or degenerative phases.

NSAID and ligament

NSAIDs have been shown to have a negative impact on collagen formation, fibroblast proliferation and healing in animal models of ligament injury.²⁹²⁻²⁹⁵ Clinical studies of ligament injuries also do not support their use. One study of patients who underwent ACL reconstruction found increased joint laxity at six weeks in an NSAID-treated group compared with a group that had not been prescribed NSAIDs.²⁹⁶ Another study demonstrated similar analgesic efficacy for paracetamol and NSAID (diclofenac) following acute ankle ligament injuries, but found increased oedema at day three in the NSAID group.²⁹⁷

NSAID and bone

Prostaglandins have an important role in bone metabolism, stimulating both osteoblasts and osteoclasts.²⁹⁸ Numerous animal studies have reported a delay in bone healing with NSAID treatment.²⁹⁹ It appears that COX-2 inhibition has the most negative impact due to inhibition of endochondral ossification.³⁰⁰⁻³⁰² Although high-quality clinical studies are lacking, a number of studies suggest a delay in bone healing with NSAID use.^{303, 304}

PRACTICE PEARL

NSAIDs could be considered a risk factor for fracture healing and cannot be recommended in fracture management.

Adverse effects of NSAIDs

Adverse effects of NSAIDs may be associated with the gastrointestinal, cardiovascular and renal systems, and susceptibility increases with prolonged use and patient age.³⁰⁵ The most common adverse effects are gastrointestinal with epigastric pain, nausea or reflux symptoms. The risk of experiencing these dyspeptic adverse effects can be lowered by using the minimum effective dose, taking the medication with food, using

gastroprotective formulations or the concomitant use of proton pump inhibitors (such as omeprazole or lansoprazole) or H2 antagonists (such as ranitidine). Occult gastrointestinal bleeding may contribute to iron depletion and therefore clinicians should enquire about NSAID self-prescribing in iron-deficient athletes.

Selective COX-2 inhibitors with reduced gastrointestinal side effects have been available since 1999. In 2004, a number of COX-2 inhibitors (valdecoxib and rofecoxib) were taken off the market, as they were specifically associated with a high number of adverse vascular events. COX-2 inhibitors, and some non-selective NSAIDs such as diclofenac, are associated with an increased risk of thrombotic events, and are contraindicated in ischaemic heart and cerebrovascular disease. They should be used with caution in patients with risk factors for cardiovascular events. However, they are often used in athletes with musculoskeletal injury due to their effective analgesic properties and low adverse effect profile.

Medications for neuropathic pain and central sensitisation

Neuropathic pain is defined as pain arising from a lesion or disease affecting the somatosensory nervous system (Chapter 6).^{306, 307} It is often characterised clinically by amplified pain responses following noxious or non-noxious stimuli and may be present in combination with true nociceptive pain arising from tissue damage. In patients with neuropathic pain, both peripheral and central neural sensitising mechanisms may be present.

Recent research has suggested that central sensitisation is a potentially important feature of a number of musculoskeletal presentations, including shoulder pain, lateral epicondylalgia and patellar tendinopathy.^{308, 309} Central sensitisation is defined as 'an amplification of neural signalling within the central nervous system (CNS) that elicits pain sensitivity'.³¹⁰ It is an important feature for clinicians to detect and treat within the overall rehabilitation plan.

The management of neuropathic pain or central sensitisation in musculoskeletal injury should have a number of interdisciplinary elements and may incorporate therapeutic medication. In addition to the analgesic medication previously discussed, a number of other pharmaceuticals may have a therapeutic role. However, although these medications have demonstrated efficacy in RCTs and meta-analyses, there have been no clear predictors of treatment success in patients with neuropathic pain. The therapeutic approach is therefore usually a stepwise trial approach to identify which medication or combination provides the greatest pain relief with the fewest side effects.

Tricyclic antidepressants

Tricyclic antidepressants inhibit serotonin and nor-adrenaline neurotransmitter re-uptake and block sodium channels. Nortriptyline is most widely used and is the active metabolite of amitriptyline. It is recommended in UK, European, US and Australian guidelines for the treatment of neuropathic pain, although not as a first-line treatment.³¹¹ A dose of 10-25 mg is usually prescribed at night to avoid sedative effects during the day. A recent Cochrane review concluded that there was little evidence to support the use of nortriptyline in neuropathic pain, particularly when compared with duloxetine and pregabalin.³¹¹

Serotonin reuptake inhibitors

Duloxetine is a serotonin re-uptake inhibitor with moderate to strong evidence for efficacy in the treatment of neuropathic pain.^{312, 313} A starting dose of 30 mg is often increased to an efficacious treatment dose of 60 mg. Similar to other medications for neuropathic pain, fatigue may be a side effect.

Gabapentin and pregabalin

Gabapentin and pregabalin are gamma-aminobutyric acid analogues that bind to calcium channels on afferent nociceptors and subsequently reduce neurotransmitter release. Pregabalin has a half-life of around six hours.³¹⁴ Starting dose of pregabalin is usually 75 mg twice daily and this may be titrated to a total daily dose of 300 mg over the course of one week. The most common adverse effect of these medications is fatigue or a mild sedative effect, and they are often not appropriate to take before training sessions. Gabapentin is also available in an extended-release preparation that may be more effective for neuropathic pain, but is more likely to impact on athlete's training programs.

Local anaesthetic patches

Topical lignocaine (lidocaine) patches have demonstrated positive effects in short-term studies on neuropathic pain³¹² and may be helpful as a second-line treatment, although the quality of the evidence is weak.

Counterirritants

Topical heat rubs are often used by athletes and mainly act as counterirritants. Their mechanism of action is probably related to their effect on transient receptor potential (TRP) channels (Fig. 17.22). These channels are sensitive to heat and initiate a local inflammatory response. Prolonged activation of the nociceptors may result in a depletion of local neurotransmitters, which may result in a reduction in pain and nociceptor transmission from the site of the more long-standing injury. Counterirritants, such as

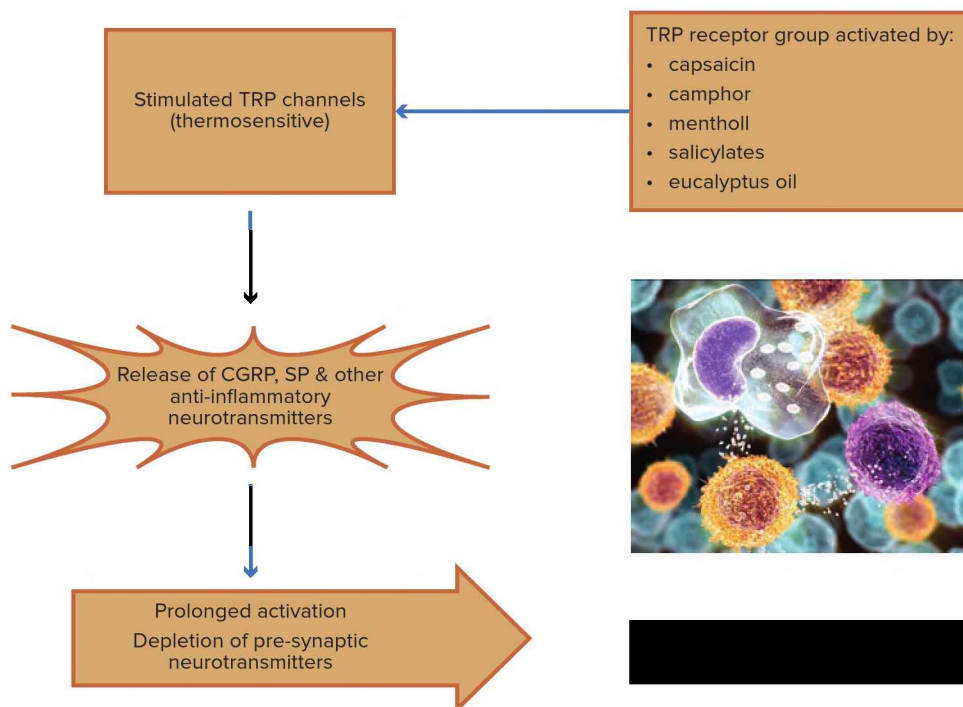


Figure 17.22 Mechanisms of action of counterirritants

Injection therapy: should injections be performed under ultrasound guidance?

There is consistent high-quality evidence that ultrasound-guided injections are more accurate than landmark-guided injections.³¹⁵ There is moderate limited-quality evidence that ultrasound-guided injections are more efficacious than landmark-guided injections, particularly in large joints, subacromial bursal injections,³¹⁶ carpal tunnel and inflamed joints.³¹⁷ The interpretation of the research in this area is challenging, as corticosteroid injection is often studied. Not only may this be an inappropriate therapeutic choice for the particular pathology,^{318, 319} but there are known systemic effects of corticosteroid injections that may affect outcome measures. It is also reasonable to assume that with the development of injectable therapeutic agents without systemic effects, such as hyaluronic acid, these would be unlikely to demonstrate maximal therapeutic efficacy unless accurately positioned. Therefore, we recommend that clinicians who regularly perform musculoskeletal injections undertake training in ultrasonography and perform injection therapy under ultrasound guidance.



Figure 17.23 Ultrasound-guided injection
USED WITH PERMISSION OF GE HEALTHCARE

capsaicin or camphor, may therefore have a use for mild analgesia before activity, but also potentially in a chronic injury that demonstrates increased neural sensitisation or elements of neuropathic pain.³¹²

Local anaesthetic injections

Local anaesthetic injections may be used in sports medicine for a number of reasons. They can be used in the treatment of musculoskeletal injury to reduce pain or

facilitate movement. Local anaesthesia may be performed on competition day to enable athletes to participate in sport despite injury. Local anaesthetic injections are also frequently used in conjunction with a corticosteroid injection to reduce needle pain.

Local anaesthetics cause a reversible block to conduction along nerve fibres by inhibiting sodium channels, particularly in small neurones. There is a wide variation in speed of onset, duration of action and potency. Lignocaine (lidocaine) and bupivacaine are the most commonly used preparations for musculoskeletal injection. Lignocaine 1% is a quick-acting local anaesthetic that is effectively absorbed from mucous membranes. The duration of anaesthesia is usually 60–120 minutes. Bupivacaine has a slower onset, taking up to 30 minutes for effect, and a longer duration of action, lasting for several hours.

Local anaesthetic in competition

Anaesthetic injections are sometimes used to enable participation in elite sport despite the presence of AC joint injuries, finger and rib injuries, and iliac crest haematomas.³²⁰ These injections appear to be relatively safe in the short and long term. We do not recommend intra-articular injections in weight-bearing joints or around tendons of the lower limb, due to the significant risk of further injury to these structures.

Local anaesthetic in routine musculoskeletal injections

Clinicians have traditionally used local anaesthetic in most musculoskeletal injection protocols to reduce the pain of the procedure. However, as local anaesthetic has been shown to be myotoxic,³²¹ chondrotoxic³²² and tendon toxic,^{323, 324} clinicians should reconsider their use of local anaesthesia during routine injection protocols.

Traumeel

Traumeel is a herbal preparation of arnica, belladonna, calendula, heparin and echinacea that is reported to have an antioxidant and anti-inflammatory action. Some clinicians inject it in the treatment of muscle injuries³²⁵ and it is also available as a topical anti-inflammatory. The topical application of traumeel has been demonstrated to be as efficacious as topical diclofenac for pain relief in low-grade acute ankle ligament injuries.³²⁶ There are no clinical studies that evaluate its efficacy in the treatment of muscle injuries.

Actovegin

Actovegin is a deproteinised ultrafiltrate of calf serum produced in Austria, a bovine spongiform encephalopathy-free country,³²⁷ and a licensed drug in Europe, China and

Russia. It is reported to contain trace elements, amino acids, electrolytes, and carbohydrate and fat metabolites. The active ingredients have not been identified. Some basic science studies suggest Actovegin has a role in modulation of inflammatory processes, but no quality clinical evidence exists to support its use in muscle injury.

Sclerosant

Early studies reported positive effects of sclerosing injections of polidocanol to areas of neovascularisation, just outside the Achilles tendon.^{328, 329} However, more recent studies have not demonstrated positive effects³³⁰ and it is currently not widely used in clinical practice.

Prolotherapy

Prolotherapy is a collective term for the injection of an irritant, usually hyperosmolar dextrose, in the treatment of chronic painful musculoskeletal conditions.³³¹ The mechanisms of action are not fully understood, but it is claimed that prolotherapy stimulates inflammation and growth factor release, initiates a proliferative response and results in desensitisation through a denervating effect. There is conflicting evidence regarding its efficacy.³³²

Mechanical and high-volume injections

The pain and pathological processes involved in tendinopathy are not fully understood. Some theories suggest that tendon adherence to the adjacent fat pad or neurovascular in-growth from the fat pad are implicated in the pain and pathogenesis of Achilles and patellar tendinopathy. An injection protocol that infiltrates a high volume (up to 40 ml) of normal saline (with a small amount of local anaesthetic, hydrocortisone or aprotinin) to the interface between the tendon and the fat pad has been described.^{333–335} While case series have demonstrated positive effects, further high-quality studies are needed.

Hyaluronic acid

Chondrocytes and synovial cells produce hyaluronic acid, which is the major constituent of synovial fluid and a component of the extracellular matrix of cartilage. Hyaluronic acid maintains the structural integrity of cartilage and assists with joint mobility and shock absorption. It can also modulate the inflammatory response through inhibition of pro-inflammatory factors, such as arachidonic acid and Interleukin-1 (IL-1), which have a degradative effect on the cartilage matrix.

Hyaluronic injection can be useful in the treatment of chondropathy and may cause an initial reduction in pain due to this anti-inflammatory effect. However, it is generally most effective in the medium and long term,³³⁶ probably through its stimulation of synovial cells to synthesise endogenous hyaluronic acid (Fig. 17.24). A meta-analysis

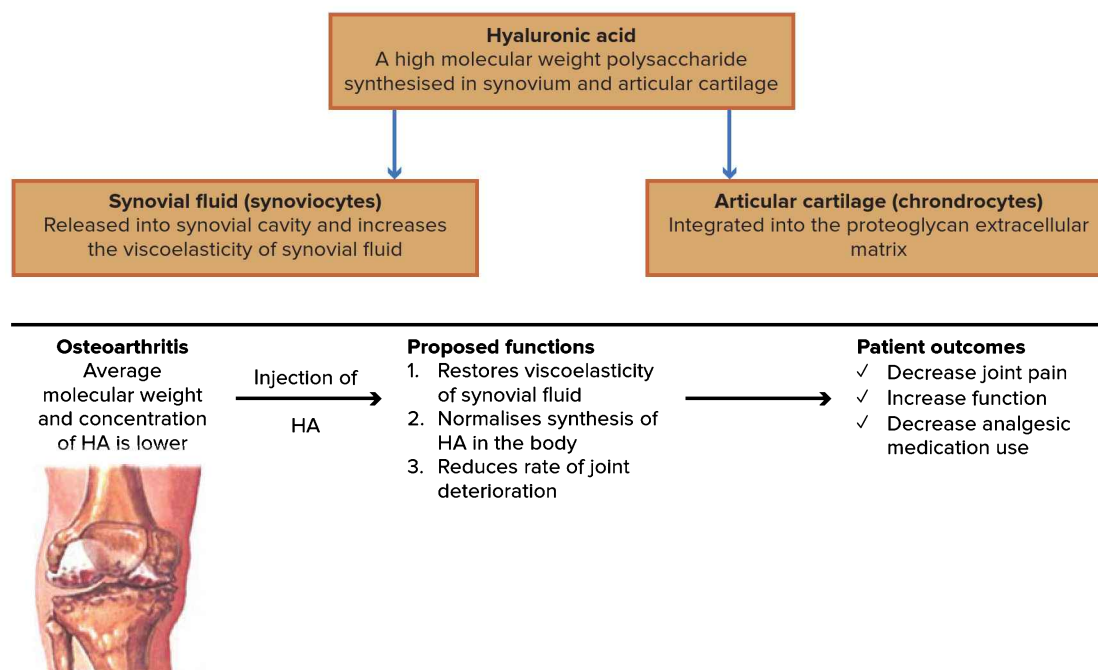


Figure 17.24 Hyaluronic acid: mechanism of action in osteoarthritis

comparing hyaluronic acid injections to corticosteroid injections suggested greater pain relief following corticosteroid at two weeks but not at four weeks, and greater benefit of hyaluronic acid at 8–26 weeks.³³⁷

Hyaluronic acid in tendons

Hyaluronic acid has been used in the field of hand surgery for several years to provide surface lubrication and enhance postoperative tendon mobility. With emerging evidence suggesting the importance of the Achilles or patellar tendon-fat pad interface in the development of tendon pathology, there have been some proposals for a role for hyaluronic acid injection in this interface³³⁸. Although there have been a number of case series publications,^{339, 340} no high-level evidence currently exists to support this therapeutic approach.

Corticosteroids

Corticosteroids are strong anti-inflammatory agents that reduce vascular permeability and leucocyte activation, and block inflammatory mediators. The proposed aim of corticosteroid injection is to reduce pain and inflammation to facilitate normal movement and allow therapeutic exercise. However, there are significant detrimental effects associated with the injection of corticosteroids. Therefore, clinicians should consider their use carefully in each case with specific regard to the current evidence, injured tissue, and the stage and goals of rehabilitation.

Peritendinous injection of corticosteroids is generally not recommended. Corticosteroids inhibit collagen synthesis and are detrimental to tendon healing. Biopsy studies have demonstrated an increase in the glutamate NMDA receptor following corticosteroid injection.³⁴¹ This receptor has been associated with chronic pain and tendinopathy.

Recent studies have demonstrated negative medium- and long-term outcomes from corticosteroid injection around elbow common extensor origin tendinopathy (tennis elbow) even compared with a 'wait and see' approach.³¹⁸ However, there is no doubt that they can provide effective pain relief in the short term and, in certain cases after informed discussions with the athlete and coach, this may be the main priority. Corticosteroid injection may be of value in inflammatory bursitis with healthy local tendon tissue. However, even in apparently isolated retrocalcaneal bursitis, caution is advised in elite athletes who place high demands on their Achilles tendons.

Intra-articular injections of corticosteroid for joint disease will reduce synovial inflammation and catabolic enzymes that are probably implicated in cartilage breakdown. However, there are negative joint consequences to be considered with respect to intra-articular corticosteroid injection. Cartilage matrix degradation and reduction in elasticity has been observed following intra-articular injections of corticosteroids to weight bearing joints.

Corticosteroid, particularly if combined with local anaesthetic, is known to be toxic to chondrocytes,^{342, 343}

and repeated injections may limit cartilage repair and regeneration. The injection of corticosteroid followed by a quick return to running is thought to be more detrimental to the articular cartilage. Injection directed to the inflamed synovial tissue rather than the joint fluid may be considered to limit negative impact on the cartilage. In summary, corticosteroid injection into weight-bearing joints should be approached with caution and considered only in the presence of excessive inflammation. There may be a more appropriate and effective role for corticosteroid injection into smaller, nonweight-bearing joints or spinal, apophyseal joints or epidural injections.

Other adverse effects of corticosteroids

The adverse effects of corticosteroids on tendons and cartilage have been presented above. However, there are other side effects that should be considered and discussed with the patient. Corticosteroid injection can commonly result in skin depigmentation, which may be permanent. Fat atrophy is also a possible effect, particularly in superficial injections. The risk of this can be minimised by accurate guided-needle placement and care to avoid steroid placement along the needle track during withdrawal.

Intra-articular injection may result in pericapsular calcification and severe adverse events such as avascular necrosis have been reported, usually following several injections within a short time frame. Joint infection is another significant serious adverse effect and appropriate sterilisation measures with a good injection technique are essential for these injections. The presence of an overlying skin infection is a contraindication to injection.

Corticosteroid injections commonly cause a short-term exacerbation in symptoms for several days after the injection. Patients are recommended to have a short period of rest and to modify athletic activity after a corticosteroid injection.

Choice of corticosteroid

The most commonly used corticosteroids are triamcinolone, methylprednisolone, hydrocortisone and dexamethasone. They each have a different duration of action and solubility that should be considered when choosing an appropriate steroid for injection.

Triamcinolone and methylprednisolone are hydrophobic preparations and therefore form microcrystalline suspensions. As a result, they should have a longer duration of action than hydrophilic preparations, such as dexamethasone or hydrocortisone, which have a quicker onset of action but reduced duration of effect. Triamcinolone and methylprednisolone are particulate steroids that, according to recent guidelines in the USA, should be avoided in spinal injections. This is due to the risk of inadvertent arterial puncture and subsequent

particulate injection with severe vascular consequences. The debate on particulate versus non-particulate spinal injections continues, but there is some evidence to suggest that non-particulate steroids, such as dexamethasone, have an equivalent effect. Therefore, it may be prudent to choose this form of steroid for spinal injections.^{344, 345}

Other medications

Sleep medication

Sleep is of great importance to the athlete in normal training to enable adaptation and recovery (Chapter 13). A lack of sleep is associated with an increased injury risk and impaired sporting performance. During injury rehabilitation, sleep can be disrupted due to pain and anxiety. For the injured athlete, good quality sleep is essential for many reasons, but particularly to enable optimal protein synthesis, for soft tissue healing and strengthening, and neuromotor learning to achieve the gait, running or skill re-education required during rehabilitation. The sleep strategy for an athlete in rehabilitation is a broad remit for the multidisciplinary team, but medication may have a useful short-term role.

There are several medications that can assist with sleep. The clinician should be aware of the dependence and tolerance that can occur with the use of hypnotics and should therefore reserve their use for short courses (of several days only). The use of sleep medication should be carefully documented in the patient's notes, particularly if there are a number of team physicians. Generally, short-acting hypnotics are the most appropriate for athletes with sleep disturbance associated with acute injury. These will assist with sleep onset and not have a sedative effect the following day.

Temazepam is a short acting benzodiazepine with little 'hangover' effect, although tolerance can develop quickly. Diazepam is a long-acting benzodiazepine that also reduces muscle spasm and can be of use if this is also a therapeutic goal, most commonly in the treatment of acute low back pain with muscle spasm. Non-benzodiazepine hypnotics, such as zolpidem or zopiclone, have a short duration of action without hangover effects.

PRACTICE PEARL

Clinicians, as well as athletes, should always be aware of the dangers involved with the abuse of sleep medications. There have been multiple high-profile cases of such abuse across a multitude of sports. When athletes are prescribed sleep medications, they should always be warned of interactions with other drugs, including alcohol, which if taken in combination can cause enhanced central nervous system depression and lead to catastrophic consequences.

Quinolone antibiotics

Unless there is no suitable antibiotic alternative, quinolone antibiotics (e.g. ciprofloxacin, ofloxacin) should not be prescribed to athletes. There is clear evidence linking quinolones with both acute tendon rupture and increased lifetime risk of tendon disease.^{346, 347}

Glyceryl trinitrate patches

Glyceryl trinitrate (GTN) patches have been proposed to modulate tendon healing by providing nitric oxide, which may stimulate collagen synthesis. There were some early studies that demonstrated improvements in tendon pain and function in patients with non-insertional Achilles, supraspinatus and elbow common extensor origin tendinopathies.³⁴⁸⁻³⁵⁰ However, more recent RCTs fail to demonstrate clinical improvements compared with, or as an adjunct to, exercise rehabilitation.³⁵¹⁻³⁵⁴ There is a notable incidence of headache associated with initiation of treatment. Current evidence suggests that GTN patches do not appear to have an important therapeutic role in the management of tendinopathy.

Bisphosphonates

Bisphosphonates inhibit osteoclastic function and therefore increase bone mass. They have been used for many years in the treatment of osteoporosis and decrease fracture risk in this population. The use of bisphosphonates in the prevention or treatment of stress fractures and one-off intravenous injections to reduce bone pain from pubic bone stress (osteitis pubis)³⁵⁵ in athletes has been proposed.

One study found that bisphosphonates had no effect on stress fracture prevention in military recruits.³⁵⁶ In stress fracture treatment, a small study suggested bisphosphonates could result in accelerated and safe return to sport.³⁵⁷ Some clinicians also recommend the use of bisphosphonates for the treatment of athletes with osteitis pubis.

However, there are several significant concerns regarding the rationale and use for bisphosphonate treatment in athletes. Bisphosphonates have a half-life of 1-10 years, limit normal microdamage repair and suppress bone remodelling. This may be detrimental to an athlete in the long term, after recovery from the current injury, as normal bone remodelling is essential for adaptation and resilience to normal athletic training. Indeed, their short- and long-term use in an elderly osteoporotic population has demonstrated an increased risk of fatigue fractures.^{358, 359} High-quality studies of the short- and long-term impact of bisphosphonate prescription in athletes are required.

The safety of bisphosphonates in women who may subsequently become pregnant has not been established. Animal studies have shown that bisphosphonates can cross the placenta and therefore affect foetal bone

mineralisation. While there have been no case reports of teratogenic effects in humans, it would seem prudent to avoid the use of bisphosphonates in female athletes.

In addition to the potential adverse effects above, there are side effects of bisphosphonates that should be recognised. Nausea, arthralgia and myalgia for 48 hours after intravenous treatment are common. Oral bisphosphonates can induce oesophageal inflammation and erosion. Osteonecrosis of the jaw has also been reported, usually in patients over 60 years with significant dental pathology. However, this risk should be discussed and a dental examination performed prior to bisphosphonate administration.

NUTRACEUTICALS IN INJURY MANAGEMENT

with NOEL POLLOCK

In recent years there has been an increased focus on the potential role of dietary supplements and herbal products—often collectively referred to as nutraceuticals—in the treatment of sports injuries.

Glucosamine, chondroitin and omega-3 fatty acids

Glucosamine has been advocated as a treatment for cartilage injury. The availability of glucosamine is a rate-limiting step in proteoglycan production.³⁶⁰ Proteoglycans are large complex molecules which provide elasticity and integrity to cartilage tissue. There are numerous basic science studies that provide evidence for the important role of glucosamine in supporting joint health. In addition to its role in proteoglycan production, it may also have an anti-inflammatory role and limit chondrocyte apoptosis.

Some meta-analyses have concluded that long-term treatment with glucosamine can reduce pain and disease progression, and improve joint function and mobility in patients with osteoarthritis.^{361, 362} Supplementation is often recommended in European and international guidelines on the treatment of osteoarthritis; however, while there is some rationale for its use in chondroarthritic conditions, the evidence is conflicting.³⁶³ A daily oral dose of 1500 mg is most consistently recommended.

Chondroitin sulphate is a natural glycosaminoglycan and an important component of the extracellular matrix. Similarly to glucosamine, its use has been recommended in the treatment of inflammatory arthropathy including osteoarthritis and chondroarthritic.³⁶⁴

The role of glucosamine and chondroitin in the synthesis of large proteoglycans, such as aggrecan, has led some clinicians to consider their use in patients with tendinopathy.²⁶⁸ Early tendon overload is characterised by tendon swelling and aggrecan production,³⁶⁵ and therefore further increase in large molecule proteoglycan

synthesis may be detrimental. The role of glucosamine and chondroitin in proteoglycan synthesis suggest that it may be appropriate to avoid the use of these products in early, reactive tendinopathy.

Omega-3 polyunsaturated fatty acids have been used as dietary supplements in musculoskeletal injury, particularly joint injury, due to their anti-inflammatory and antioxidant effects. Reactive oxygen species are increased in arthritis and associated with cartilage degradation. Some low-level evidence suggests that omega-3 fatty acids may be helpful at reducing morning stiffness in patients with osteoarthritis.³⁶⁶

Vitamin D

Vitamin D is a secosteroid hormone with a wide range of important physiological effects for health and athletic performance. Regarding musculoskeletal injury, Vitamin D has an essential role in the maintenance of muscle strength and mass, particularly for type II muscle fibres and in optimal bone health. Correction of vitamin D deficiency can improve muscle strength and reduce the risk of stress fractures.^{367, 368} Vitamin D deficiency is prevalent in elite athletes, and screening and treatment of deficiency is recommended.³⁶⁹

Green tea/polyphenols

Green tea and other polyphenol-rich foods such as cherry juice may have an important future role in the management of musculoskeletal injury. There have been some basic science and animal studies demonstrating a positive impact on tendon and cartilage healing.^{370, 371} Clinical studies are required to determine the clinical impact of polyphenol supplements and nutrition.

AUTOLOGOUS BLOOD, BLOOD PRODUCTS AND CELL THERAPY

with ROBERT-JAN DE VOS

Autologous blood, specific blood products, and cells are increasingly used in clinical trials in the field of musculoskeletal medicine. These autologous products secrete many growth factors that are thought to have a positive effect on healing of bone, cartilage, ligament, tendon and muscle injuries. This has been based on laboratory studies showing increased cell proliferation and collagen synthesis, and stimulation of a well-ordered angiogenesis.³⁷²⁻³⁷⁴ The rationale, applications and efficacy of these treatments are summarised below.

Autologous blood injections

The main rationale behind treatment with autologous blood injections is the presence of various growth factors

in blood with a potential healing effect. Blood is usually drawn from an arm vein and injected directly into the injured part. Ultrasound guidance can be helpful for improving delivery of autologous blood to the injury site, but in superficial injuries or in presence of clear anatomical landmarks, this is not always necessary. The amount of blood injected depends on the location and type of injury, but 3 mL or 6 mL are the most commonly used volumes.³⁷⁵

This procedure has been used mainly in tendinopathies; however, a systematic review did not show efficacy of this treatment for these injuries.³⁷⁵ More recent randomised trials show efficacy of autologous blood injections in comparison with detrimental corticosteroid injections,³⁷⁶⁻³⁷⁸ but not compared with placebo or standard physical therapy.^{379, 380}

Platelet-rich plasma

Platelet-rich plasma (PRP) is the product derived when autologous whole blood is centrifuged to separate out a preparation with a very high platelet content. The preparation is rich in both plasmatic and platelet α -granule derived growth factors, as well as many thousands of other substances.³⁷²

With the preparation of a PRP injection, a larger amount of blood is withdrawn. The blood is then placed into a tube, which in turn is placed into a centrifuge, which spins many thousand times a minute, for various lengths of time depending on the protocol. At this point, the platelets have separated from the other blood components and can be aspirated for injection. Many different versions of PRP are described depending on the duration, force and number of spins. Several sub-classifications of PRP have been made, which can serve as a basis for evaluation of efficacy in clinical studies.³⁸¹

Initially, many case series showed benefit of PRP injections in several clinical conditions.³⁸² In comparison with detrimental corticosteroid injections, it was found efficacious.³⁸³ However, systematic reviews and more recent randomised studies with appropriate control groups receiving placebo injection therapies (e.g. saline) did not show efficacy of PRP in patients with Achilles tendon ruptures,³⁸⁴ anterior cruciate ligament reconstruction,³⁸⁵ arthroscopic rotator cuff repair,³⁸⁶ Achilles tendinopathy,³⁸⁷ lateral epicondylar tendinopathy,³⁸⁸ patellar tendinopathy,³⁸⁹ plantar fasciopathy³⁹⁰ or acute hamstring injuries.³⁹¹ PRP injections showed limited to moderate evidence in a systematic review on efficacy in patients with knee osteoarthritis.³⁹² More high-quality randomised studies with low risk of bias are needed for this indication.

In conclusion, there is currently insufficient evidence to advise PRP injections as a routine part of management for musculoskeletal injuries.

Cell therapy

Preclinical studies have shown the potential for cellular therapies to regenerate rather than repair the tissues. The cells can be harvested from embryos (embryonic stem cells–ESCs), bone marrow, adipose tissue (mesenchymal stem cells–MSCs) or differentiated cells from the preferred tissue (e.g. tenocytes or chondrocytes).³⁹³ Cell lines such as ESCs have superior potency and proliferative properties, but their main disadvantages are ethical considerations and potential complications including tumorigenesis. MSCs offer promise in tissue engineering due to their proliferative capacity and the secretion of growth factors. Autologous cells can be harvested from the preferred tissue and expanded, prior to re-implantation. For example, healthy tendon cells (tenocytes) can be harvested from the patellar tendon (Fig. 17.25), expanded in a laboratory and injected into a tendinopathic tendon (e.g. Achilles). A disadvantage of this technique could be the donor site morbidity.

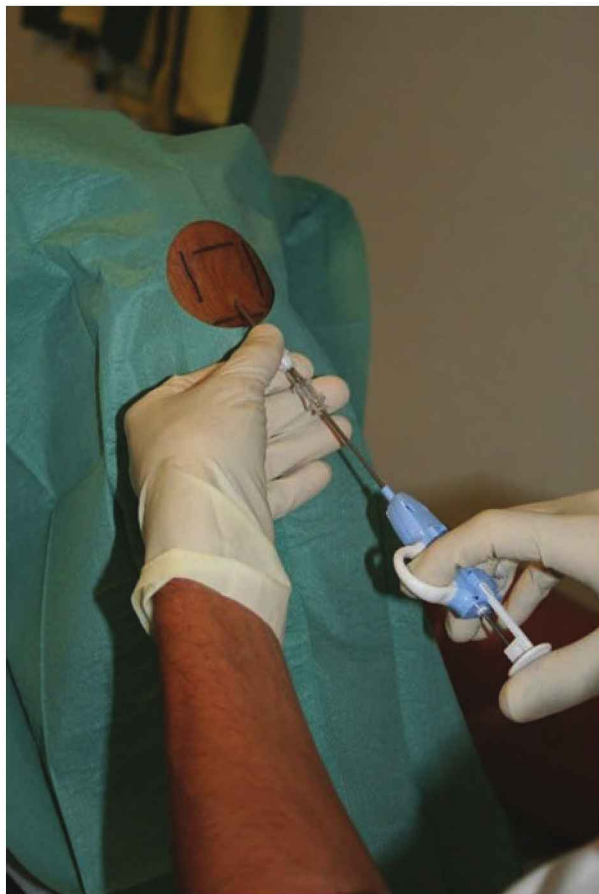


Figure 17.25 Biopsy of patellar tendon for tenocyte harvesting

Clinical studies using cell therapy for the treatment of musculoskeletal conditions are currently increasing. Small case series in patients with lateral epicondylitis and rotator cuff tendinopathy treated with autologous tenocytes, bone-marrow-derived cells and skin-derived fibroblasts showed functional and structural improvement.^{394–396} A recent randomised study in patients with patellar tendinopathy and Achilles tendinopathy showed efficacy of a local skin-derived fibroblast injection.³⁹⁷ These conclusions are, however, hampered by several major shortcomings.

Effectiveness of cell therapy is also evaluated more frequently in cartilage defects. One systematic review did not demonstrate sufficient evidence for the use of bone-marrow-derived MSCs.³⁹⁸ There is currently also insufficient evidence for surgical autologous chondrocyte implantation (ACI), the surgical implantation of healthy cartilage cells into the damaged areas.³⁹⁹ While ACI has been suggested to provide better results in tissue quality and is increasingly performed in clinical practice, more high-quality studies with long-term follow-up are needed in this field.

There are no high-quality clinical studies available to support the use of cell therapy. We expect high-quality randomised clinical trials will be published in the coming years.

In conclusion, there is currently insufficient evidence to advise autologous blood injections, PRP or autologous tendon cell injections as a routine part of management for musculoskeletal injuries.

SURGERY

Despite the many advances in the non-operative management of sports injuries, surgery has a major role to play in the management of both acute and overuse injuries. Surgery is used to remove, repair, reconstruct or realign damaged tissue. Sports surgery can be classified as arthroscopic surgery or open surgery.

Arthroscopic surgery

Arthroscopy involves the introduction of a fibre-optic telescope into a joint space to provide diagnostic information and afford the opportunity to undertake minimally invasive surgery. It is a well-established procedure for the knee, shoulder, elbow, ankle and hip, and, more recently, the wrist.

Arthroscopy utilises a light source to illuminate the joint and a video camera to capture the image, which is then displayed on one or more screens. The arthroscope is introduced through a standard portal, while another portal or portals are used to introduce operating instruments. The location of portals is important to minimise the risk

of damage to vessels and nerves. A number of instruments are available for use in arthroscopic procedures.

Arthroscopy can be carried out under local, regional or, often, general anaesthesia as a day procedure. The main areas of interest to be viewed through the arthroscope are the articular surfaces, the synovium and intra-articular structures such as the meniscus and cruciate ligaments of the knee, the glenoid labrum and rotator cuff tendons of the shoulder, and the acetabular labrum of the hip. In most joints, the majority of the articular surfaces can be viewed. Assessment of stability can be aided by combining a direct view of the joint with manoeuvres that place the joint under stress.

Common procedures performed through the arthroscope include removal of loose bodies within the joint, separation of and removal of the torn part of a meniscus, repair of a torn structure such as a peripheral tear of the meniscus or a labral detachment in the shoulder, or dividing a tight structure such as the glenohumeral joint capsule in the shoulder or scar tissue in the knee. More complex reconstructive joint procedures (ACL reconstruction, rotator cuff repair) can be performed with the aid of an arthroscope.

Arthroscopy has a low complication rate. There is a small incidence of infection and delayed portal healing. Occasionally, arthroscopy can produce a persistent joint reaction manifesting as prolonged joint effusion, persistent pain and muscle wasting. Whether this is due to the arthroscopy itself or to the underlying joint pathology is sometimes difficult to determine. Complex regional pain syndrome type 1 may occasionally develop after arthroscopy.

Open surgery

The open surgical treatment of sports-related problems includes surgery related to acute trauma and surgery for the treatment of overuse injuries. Surgery after an acute injury aims to recreate the pre-injury anatomy by the repair of damaged tissues. This may require internal bone fixation for an unstable fracture, or repair of torn ligaments or tendons. If repair of the damaged tissue is not possible, a reconstructive procedure may be performed; for example, ACL reconstruction.

Following an acute injury, the athlete may develop chronic problems, such as instability, that may require surgical repair or reconstruction. Chronic ligamentous or capsular inadequacy may develop following an injury or as a gradual process. Surgery may be required to tighten the stretched tissue, either by moving the attachment of the tissue or by a shortening procedure, such as plication, reefing or shifting.

Overuse injuries that have failed to respond to conservative measures are sometimes managed by surgical tissue release, division or excision. Excision may be performed if impingement is present or if degenerative change has led to tissue necrosis. In nerve compression, decompression or transposition of the nerve may be required. Stress fractures that fail to heal (non-union) are treated by fixation or bone graft.

With all surgical procedures, arthroscopic or open, the surgery must be considered as only a part of the treatment. Adequate post-surgical rehabilitation is as important as the procedure itself. Rehabilitation following injury and surgery is discussed in Chapter 18.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

